# Steel parts: drilling templates

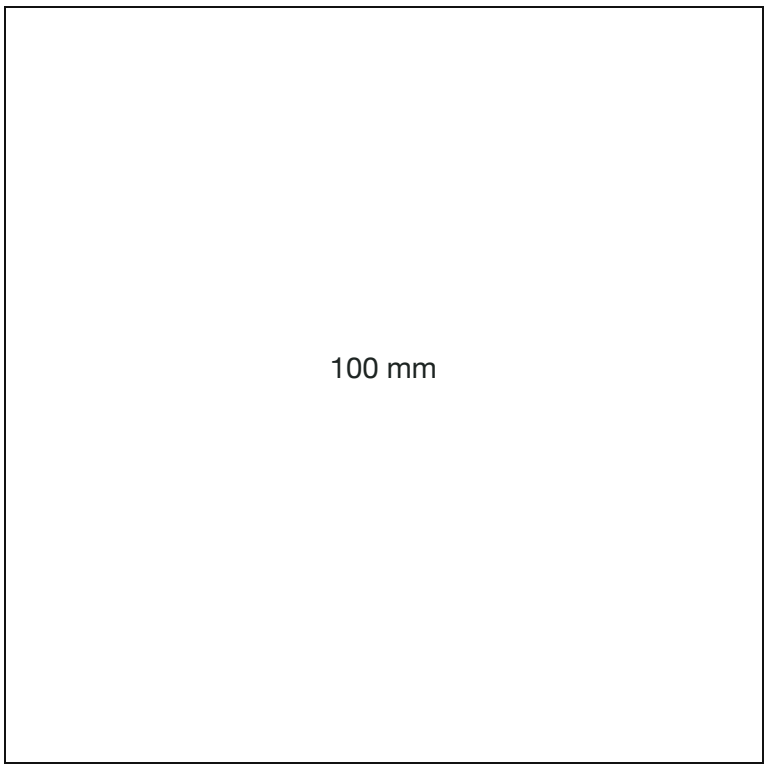
DIY CNC mill · October 2026 · full size, US Letter

## Printing

1. Print at **Actual size / 100 %**. Turn off "Fit to page" and "Scale to fit". Letter paper, portrait.
2. Measure the square below: it must be 100 mm (and the bar on every page 50 mm).

## Marking and drilling

1. Cut out the part's outline on each page and tape it to the part, its edges on the part's edges (x runs along the length from the end the page starts at). For a tube, tape it to the face named in the title.
2. Center-punch through every crosshair, pull the paper off, and drill on a drill press with the part clamped. Pilot with 1/8 in first. Tube holes marked "near wall only" go through one wall; the rest go through both.
3. Tap the holes the legend says to tap (cutting oil, a tap guide or the drill press chuck to start square).
4. Parts marked "x2" or more: mark one, then use it as a drilling jig for the rest (clamp them together).




## Pages

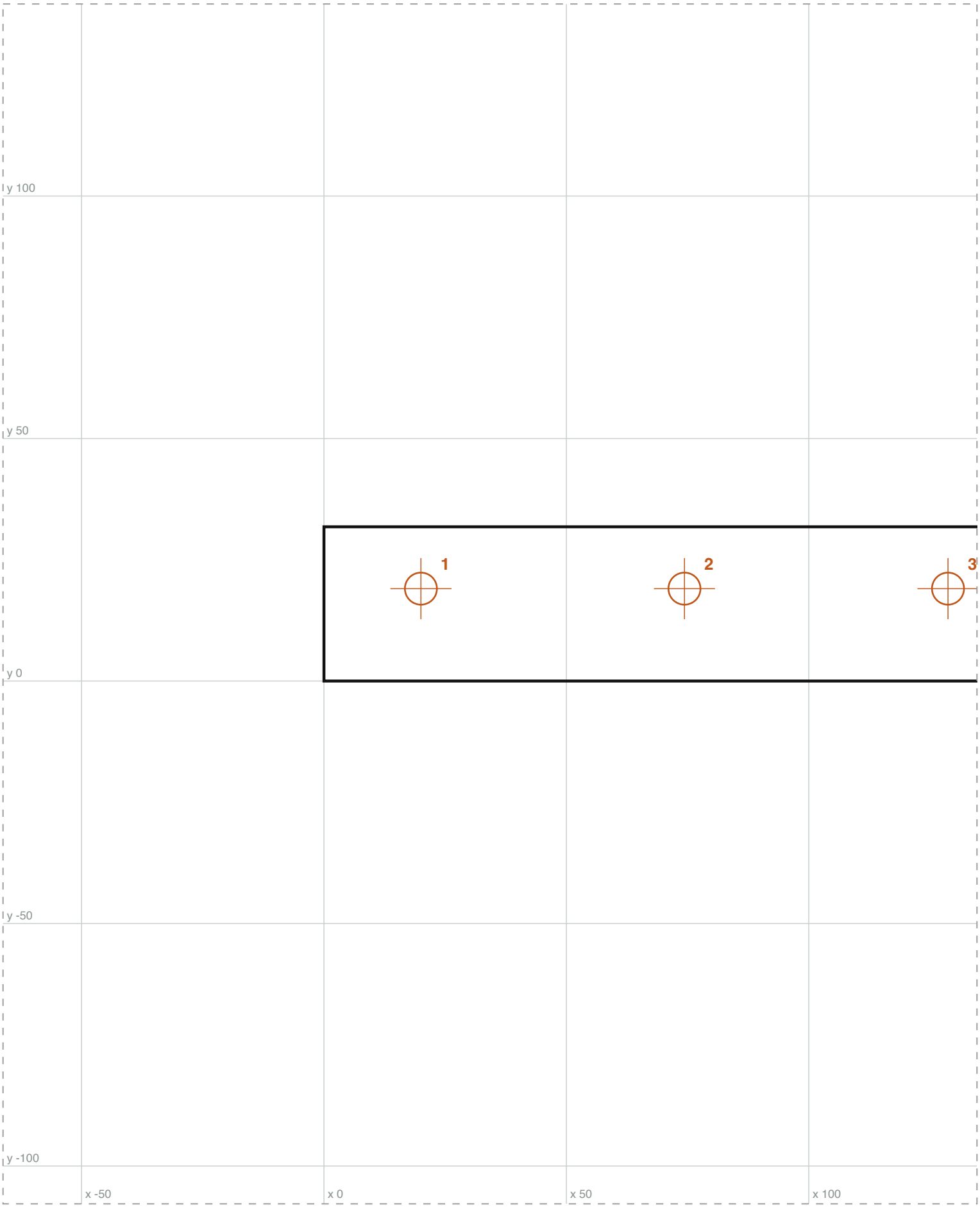
| Part | Qty | Pages | Holes |
|---|---|---|---|
| X carriage angle (flat leg, from the heel) | 2 | 2-3 (2) | 5 |
| Top plate, lower | 2 | 4-4 (1) | 7 |
| Top plate, upper | 2 | 5-5 (1) | 5 |
| Beam face plate | 1 | 6-8 (3) | 17 |
| Rail bar rear | 2 | 9-10 (2) | 9 |
| Rail bar front | 2 | 11-12 (2) | 8 |
| Upright foot | 2 | 13-14 (2) | 5 |
| Pad BF | 1 | 15-15 (1) | 4 |
| Seat rear | 2 | 16-17 (2) | 4 |
| Pedestal bar rear | 2 | 18-18 (1) | 2 |
| Beam seat | 2 | 19-19 (1) | 5 |
| Beam cap | 2 | 20-20 (1) | 4 |

# X carriage angle (flat leg, from the heel) (x2: mark each)

page A1 · full size · Angle 1-1/4 x 1-1/4 x 1/4 in, 257 mm long · lay it on the part, x along its length from one end

50 mm: measure it

6.6 mm (17/64 in)

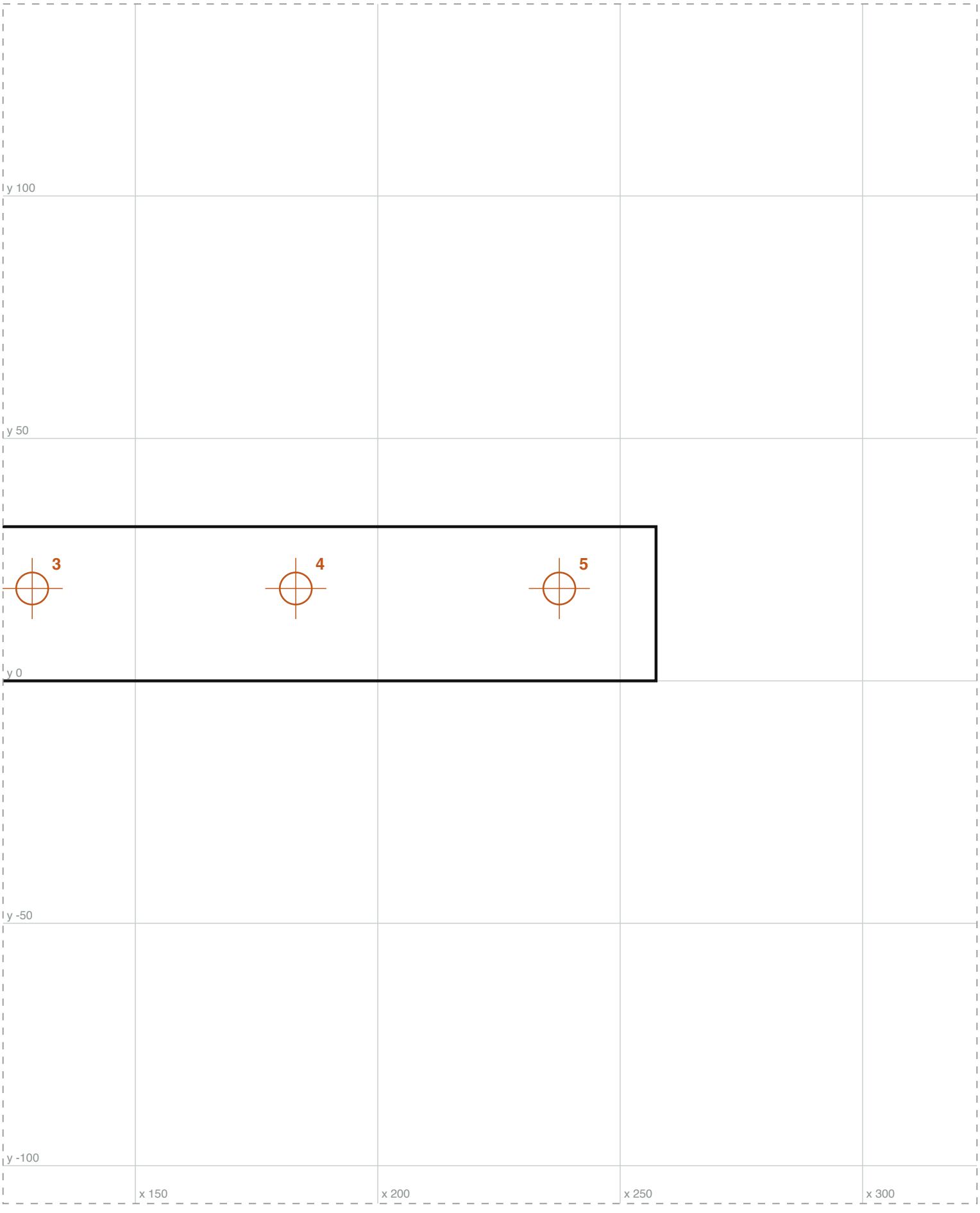


# X carriage angle (flat leg, from the heel) (x2: mark each)

page A2 · full size · Angle 1-1/4 x 1-1/4 x 1/4 in, 257 mm long · lay it on the part, x along its length from one end

50 mm: measure it

6.6 mm (17/64 in)

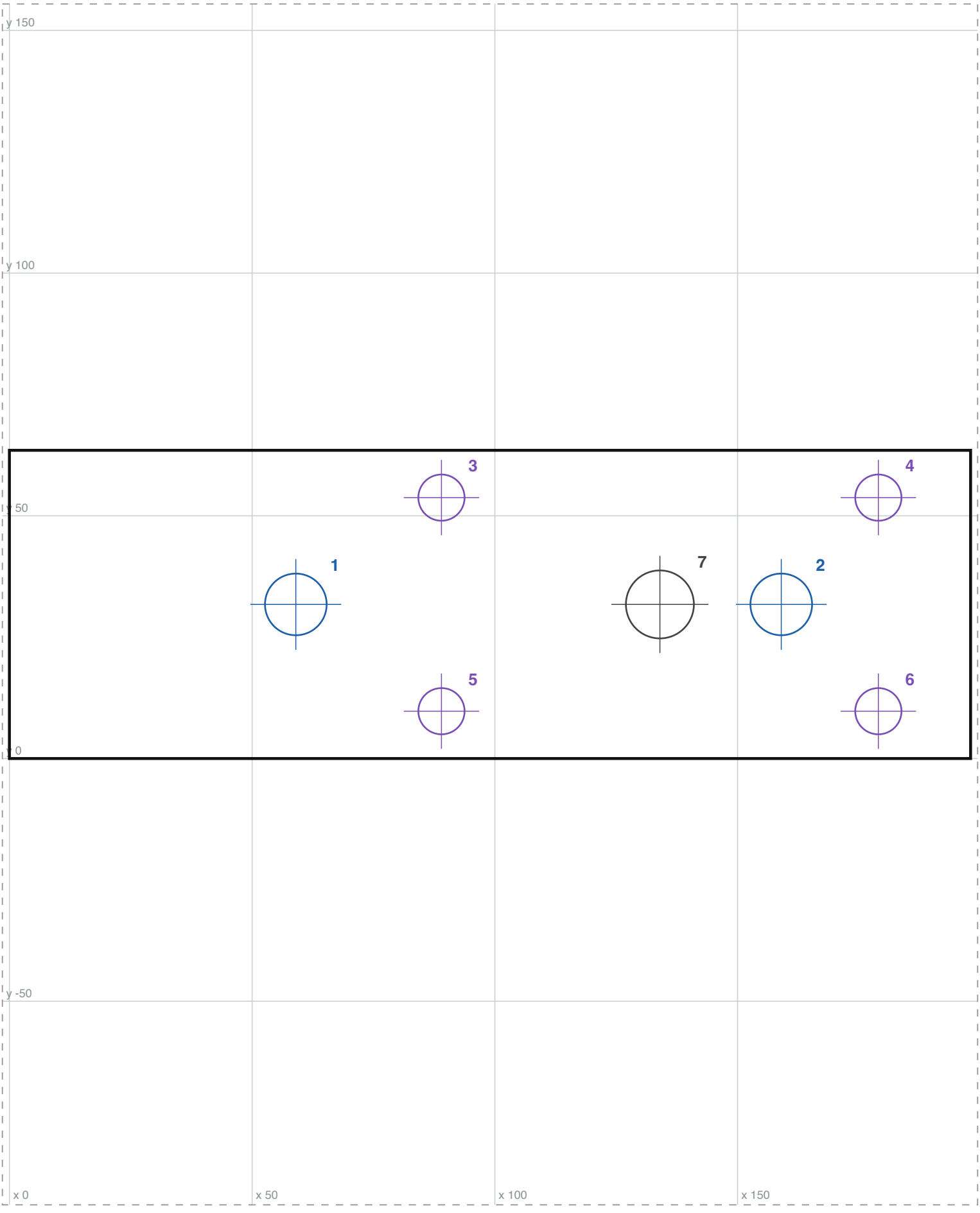


# Top plate, lower (x2: mark each)

page A1 · full size · Flat bar 1/2 x 2-1/2 in, 198 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 9.53 mm (5/16 in, tap 3/8-16) ● 12.7 mm (27/64 in, tap 1/2-13) ● 14 mm (9/16 in)

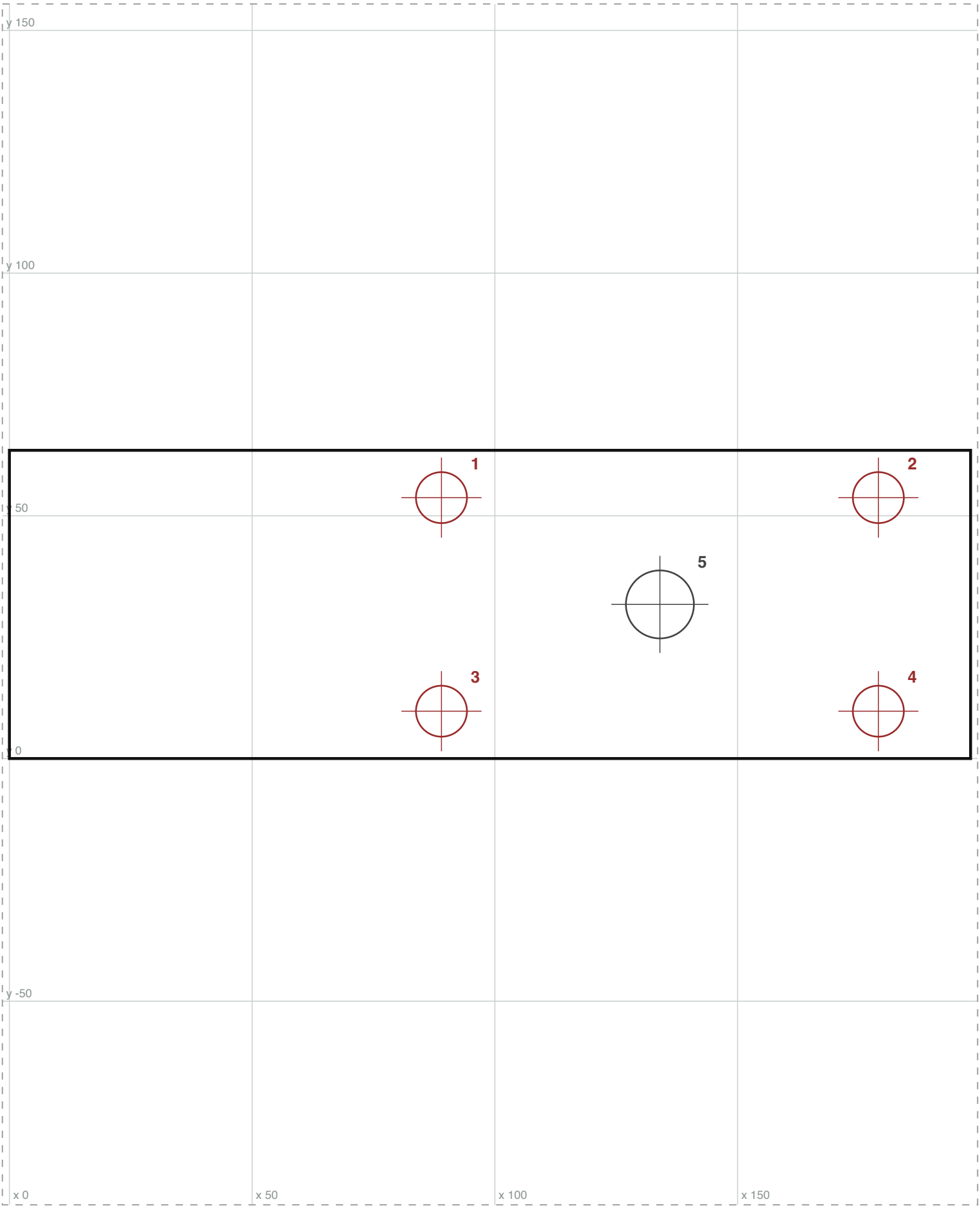


# Top plate, upper (x2: mark each)

page A1 · full size · Flat bar 1/2 x 2-1/2 in, 198 mm long · lay it on the part, x along its length from one end

50 mm: measure it

10.5 mm (13/32 in)  14 mm (9/16 in)

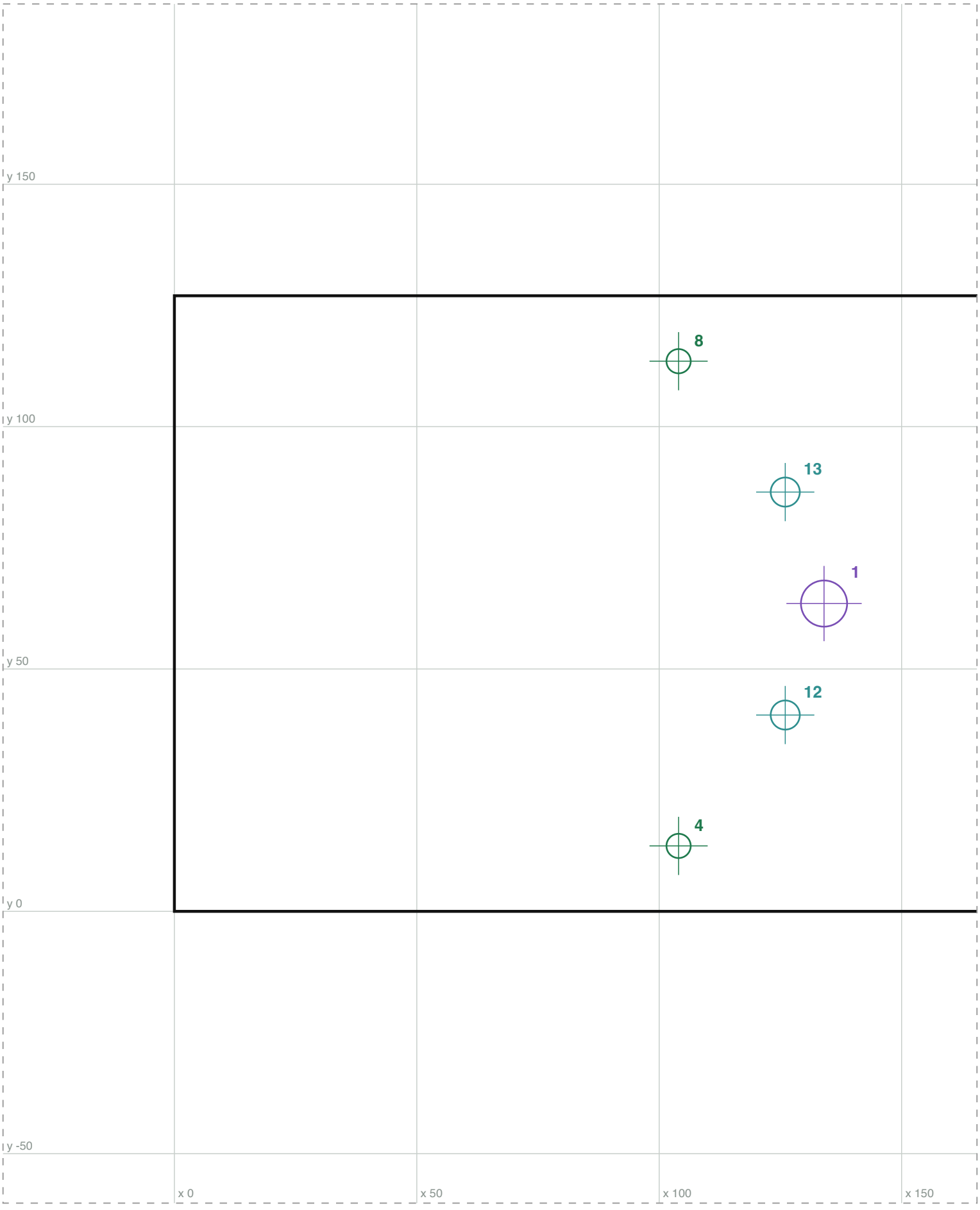


# Beam face plate

page A1 · full size · Flat bar 1/2 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 5 mm (M5 tap drill (4.2 mm), tap M5) ● 6 mm (M6 tap drill (5 mm), tap M6) ● 9.53 mm (5/16 in, tap 3/8-16)

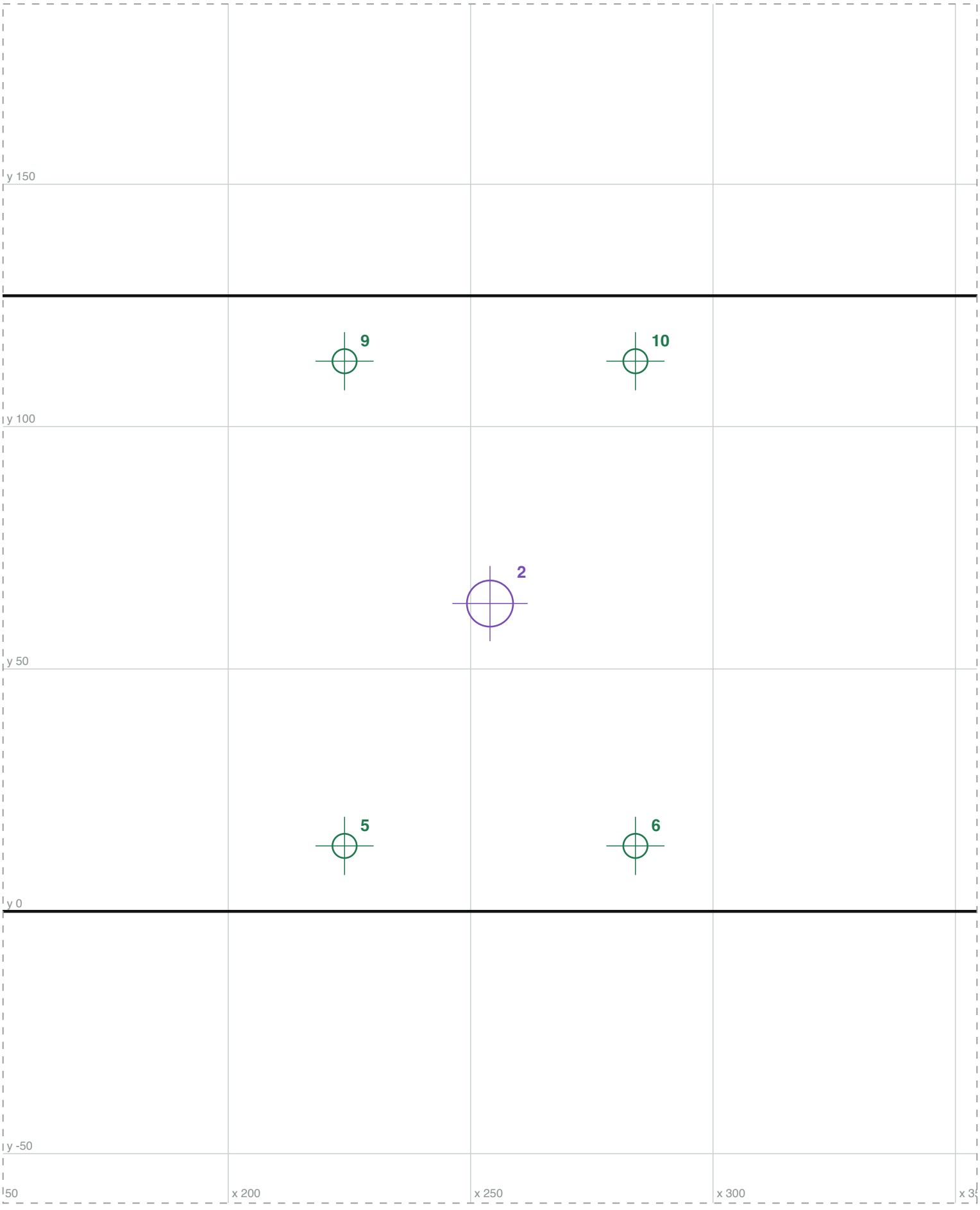


# Beam face plate

page A2 · full size · Flat bar 1/2 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 5 mm (M5 tap drill (4.2 mm), tap M5)  ● 9.53 mm (5/16 in, tap 3/8-16)

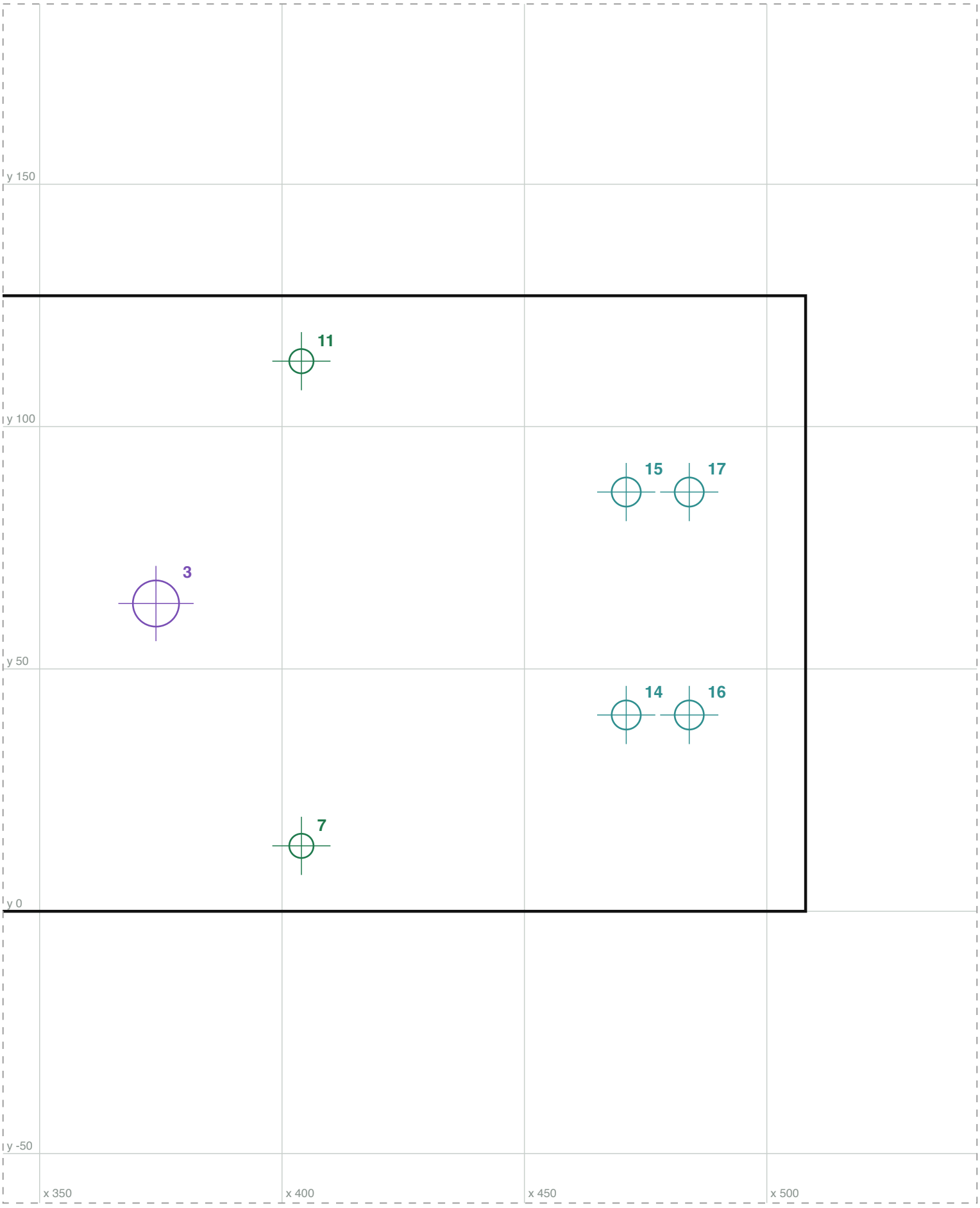


# Beam face plate

page A3 · full size · Flat bar 1/2 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 5 mm (M5 tap drill (4.2 mm), tap M5) ● 6 mm (M6 tap drill (5 mm), tap M6) ● 9.53 mm (5/16 in, tap 3/8-16)

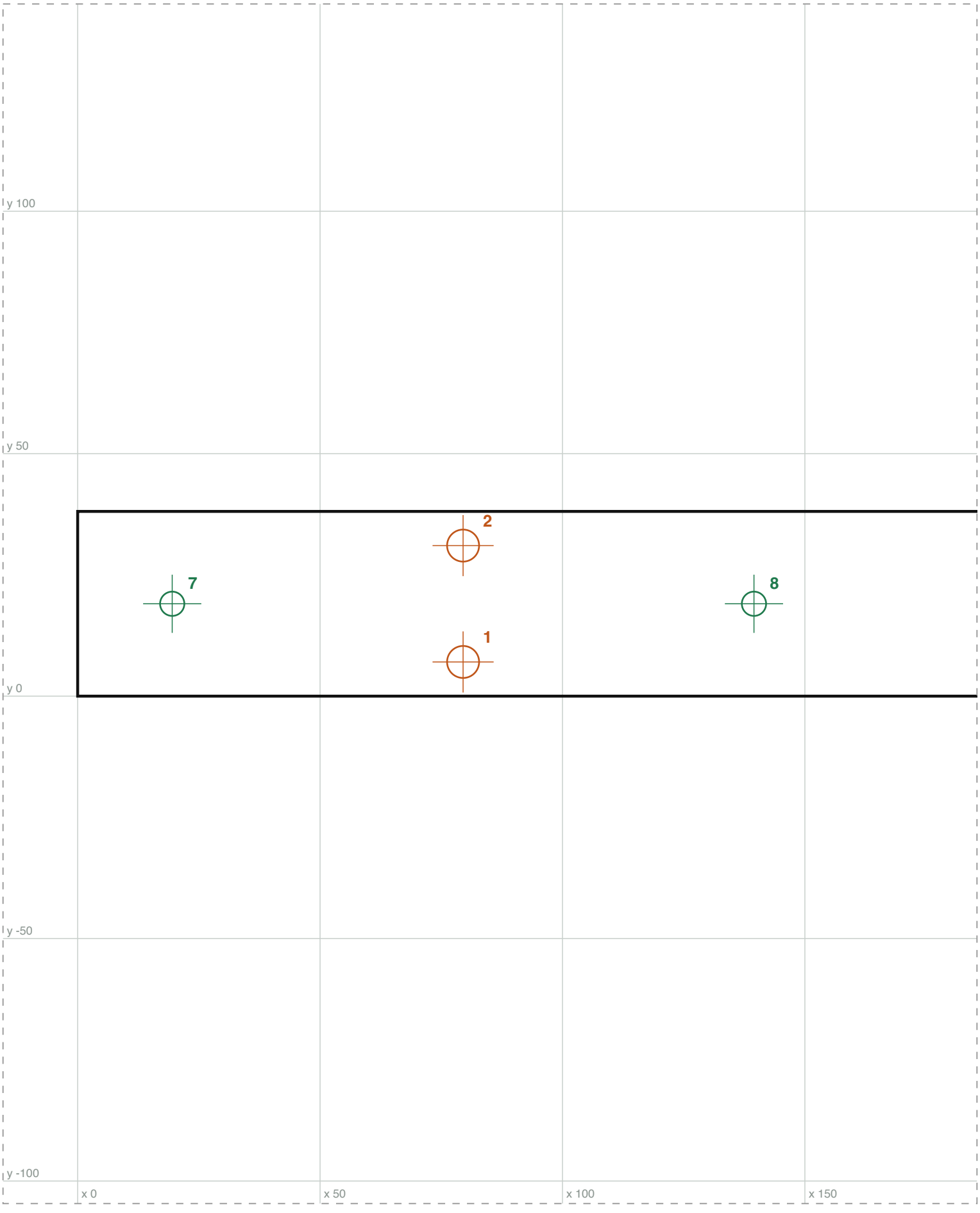


# Rail bar rear (x2: mark each)

page A1 · full size · Flat bar 3/8 x 1-1/2 in, 359 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 5 mm (M5 tap drill (4.2 mm), tap M5)  ● 6.6 mm (17/64 in)



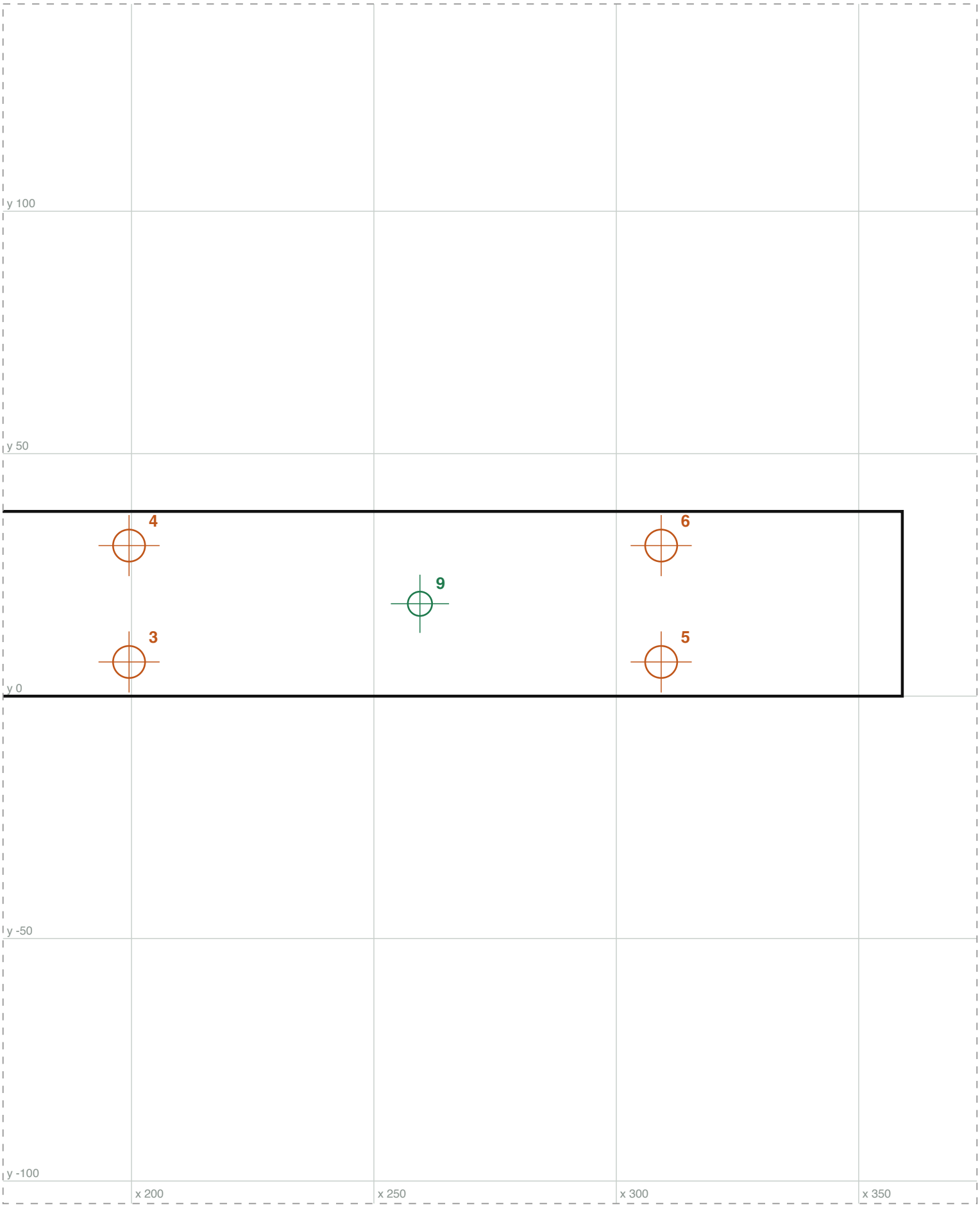
# Rail bar rear (x2: mark each)

page A2 · full size · Flat bar 3/8 x 1-1/2 in, 359 mm long · lay it on the part, x along its length from one end

50 mm: measure it

5 mm (M5 tap drill (4.2 mm), tap M5)

6.6 mm (17/64 in)



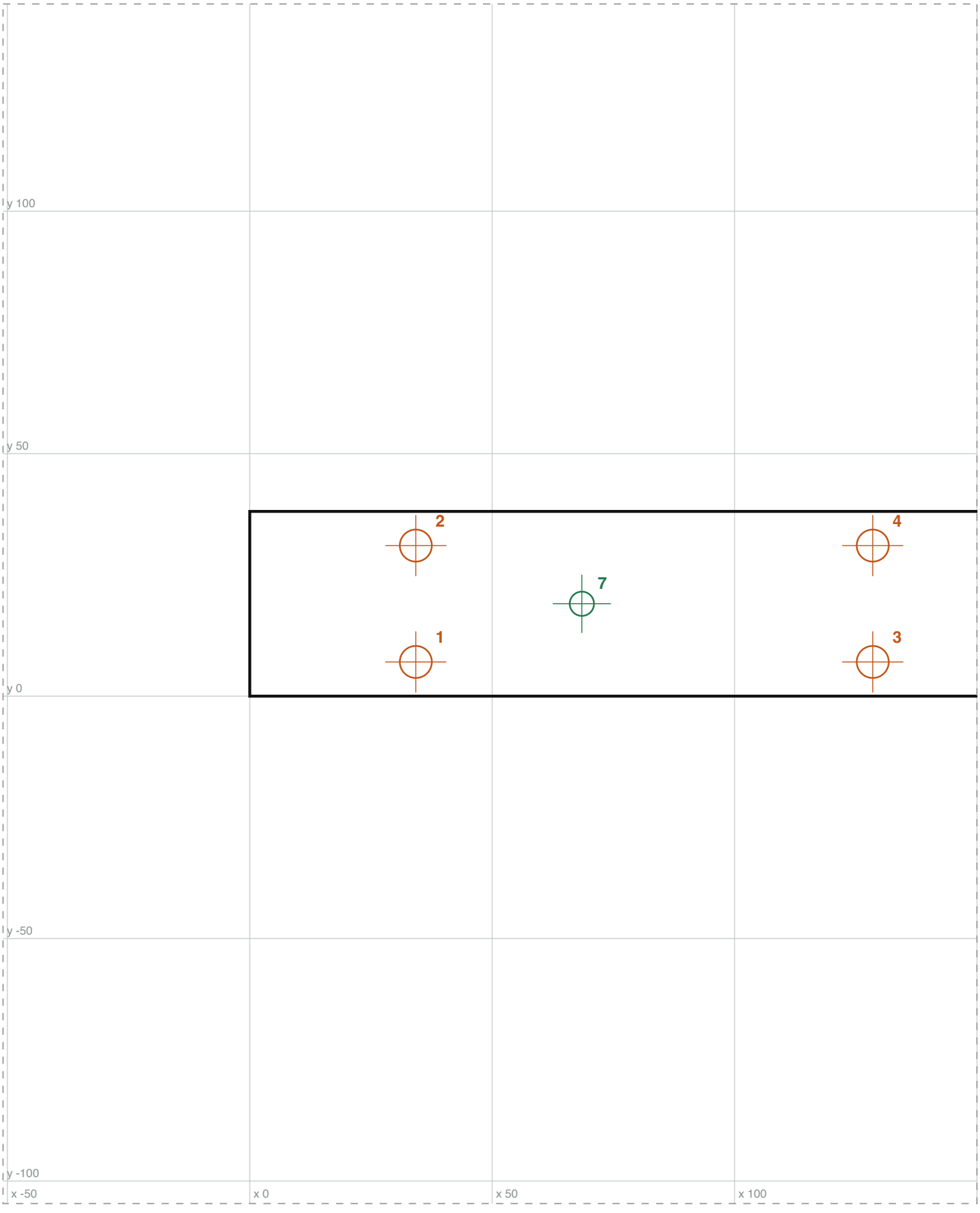
# Rail bar front (x2: mark each)

page A1 · full size · Flat bar 3/8 x 1-1/2 in, 288 mm long · lay it on the part, x along its length from one end

50 mm: measure it

5 mm (M5 tap drill (4.2 mm), tap M5)

6.6 mm (17/64 in)

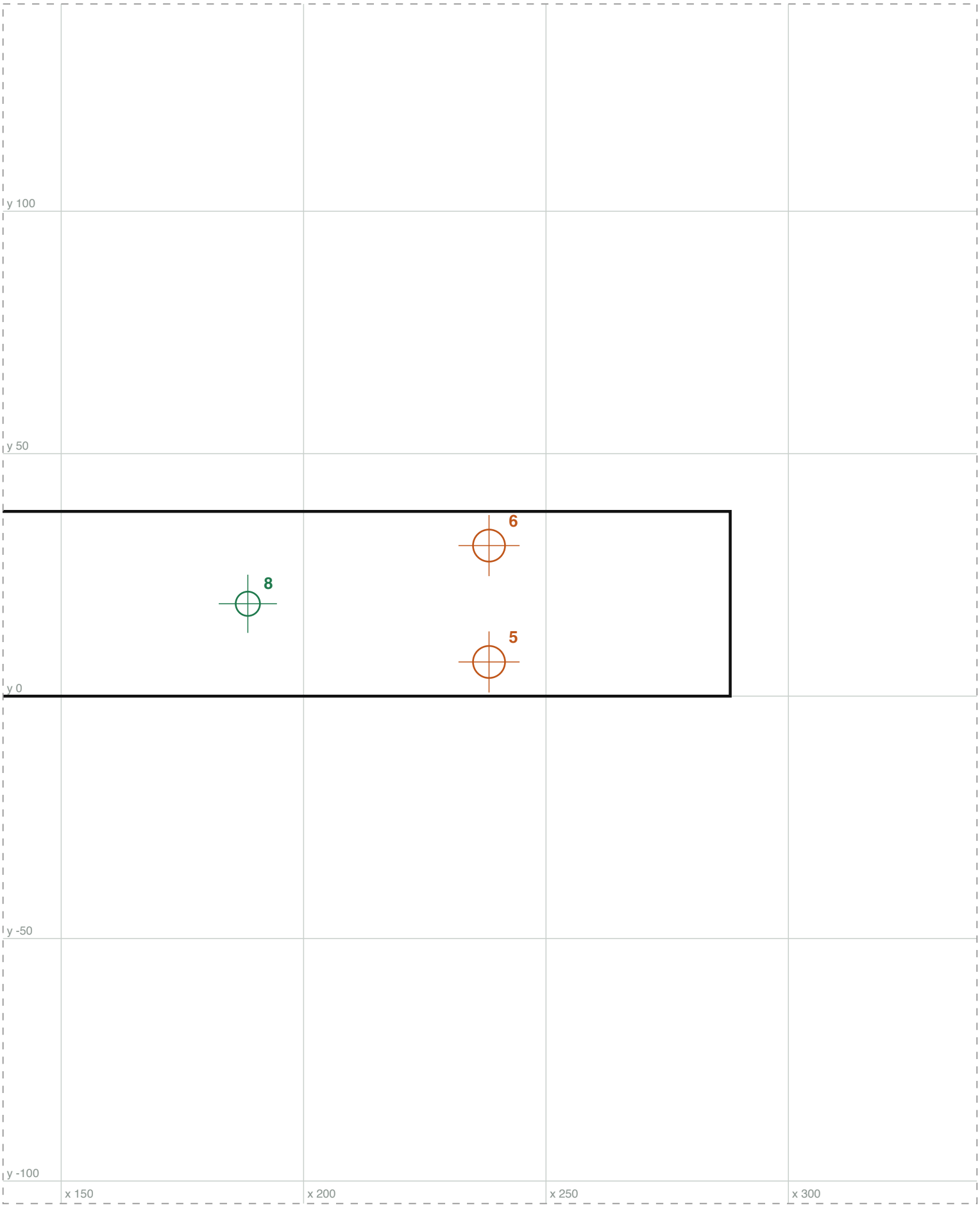


# Rail bar front (x2: mark each)

page A2 · full size · Flat bar 3/8 x 1-1/2 in, 288 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 5 mm (M5 tap drill (4.2 mm), tap M5)  ● 6.6 mm (17/64 in)

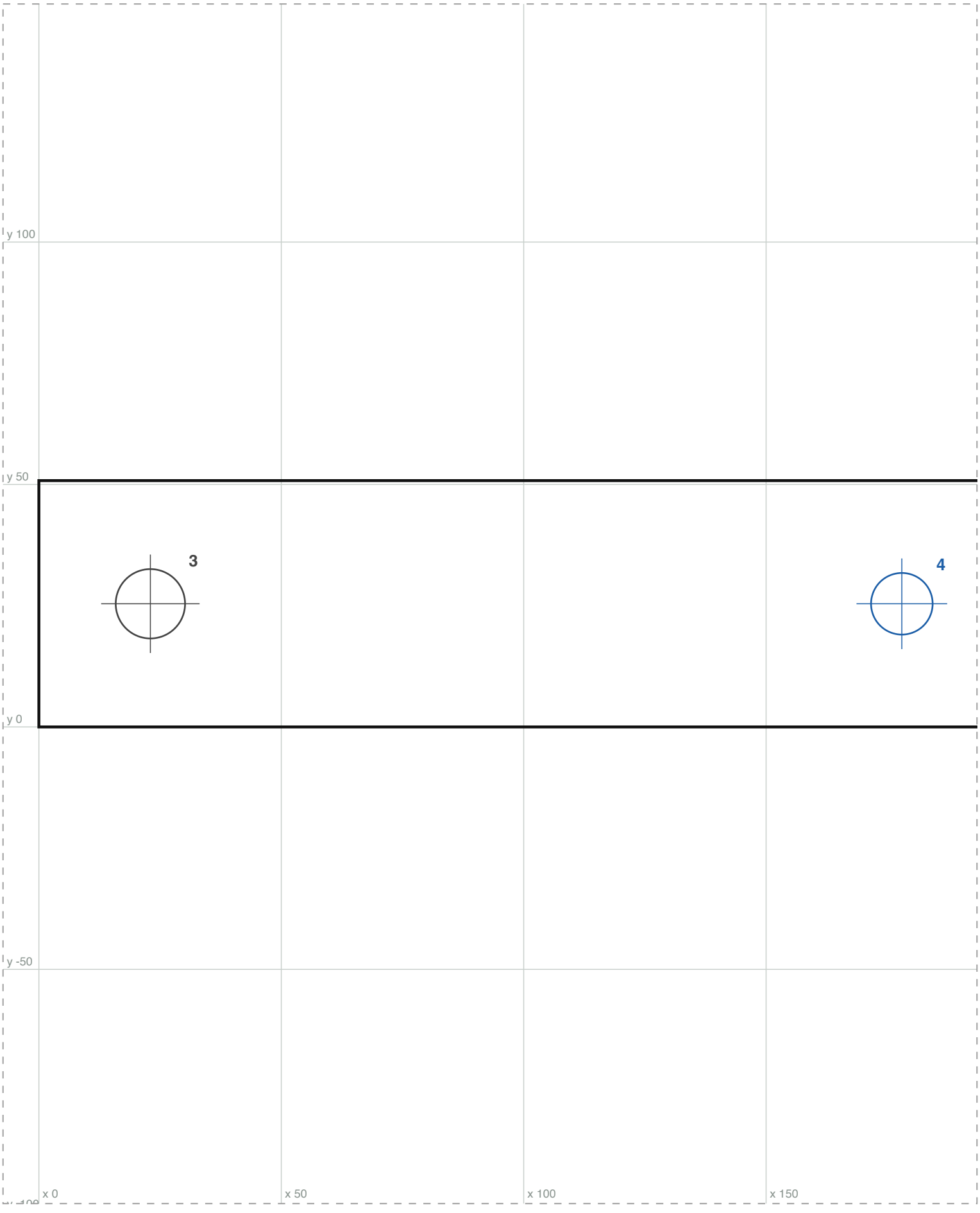


# Upright foot (x2: mark each)

page A1 · full size · Flat bar 3/8 x 2 in, 375 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 12.7 mm (1/2 in)  ● 14.3 mm (9/16 in)

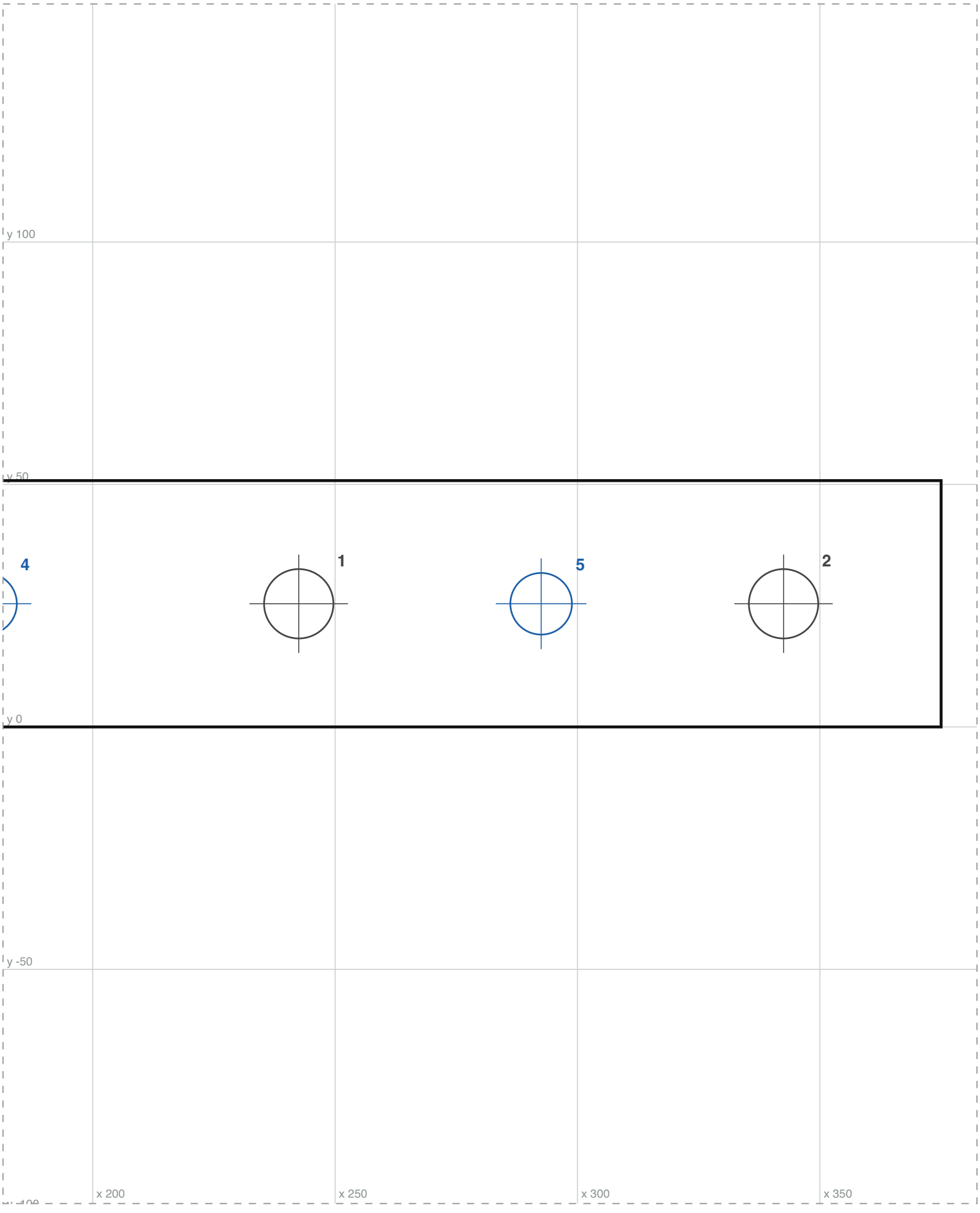


# Upright foot (x2: mark each)

page A2 · full size · Flat bar 3/8 x 2 in, 375 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 12.7 mm (1/2 in)  ● 14.3 mm (9/16 in)

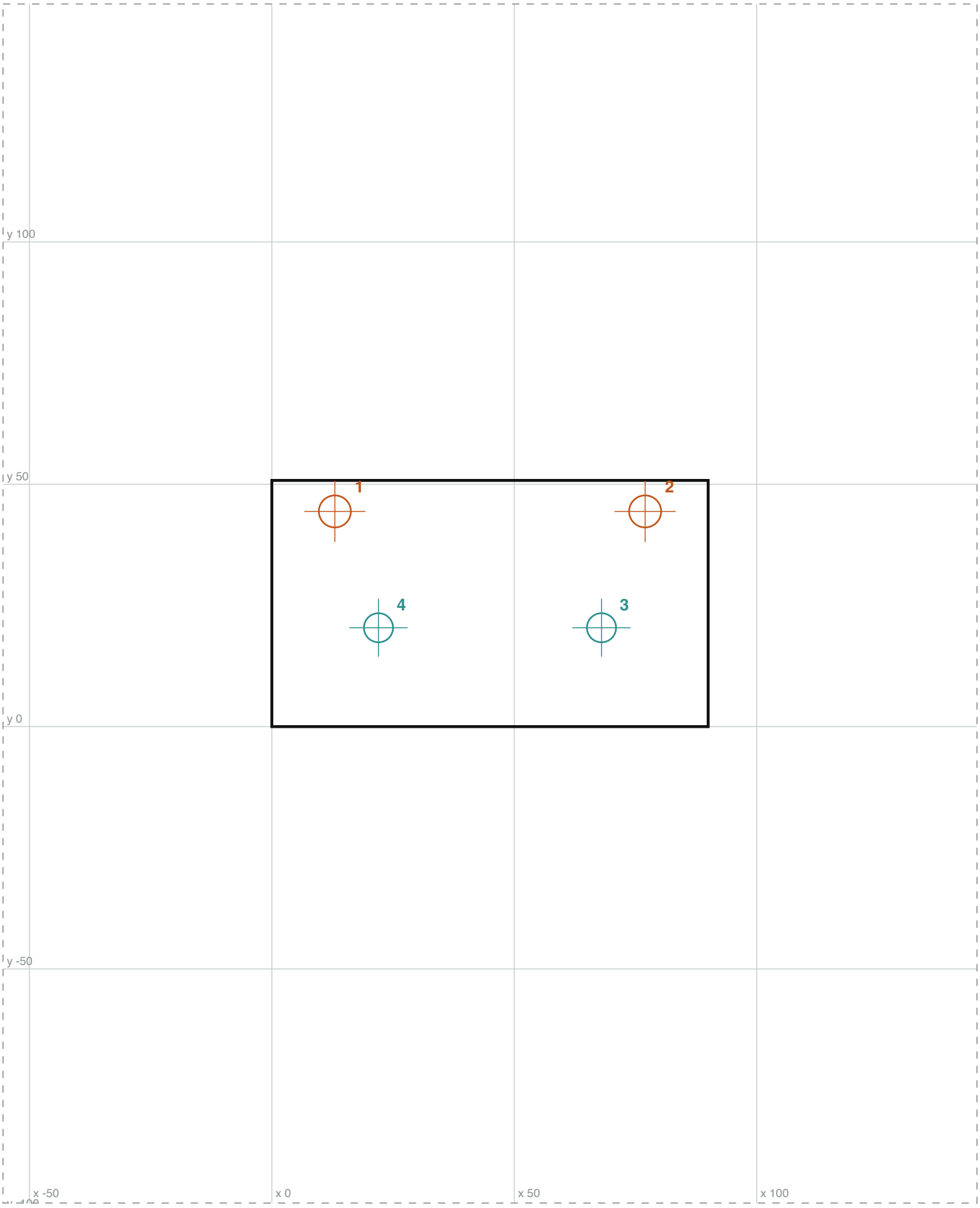


# Pad BF

page A1 · full size · Flat bar 3/8 x 2 in, 90 mm long · lay it on the part, x along its length from one end

50 mm: measure it

6 mm (M6 tap drill (5 mm), tap M6)  6.6 mm (17/64 in)

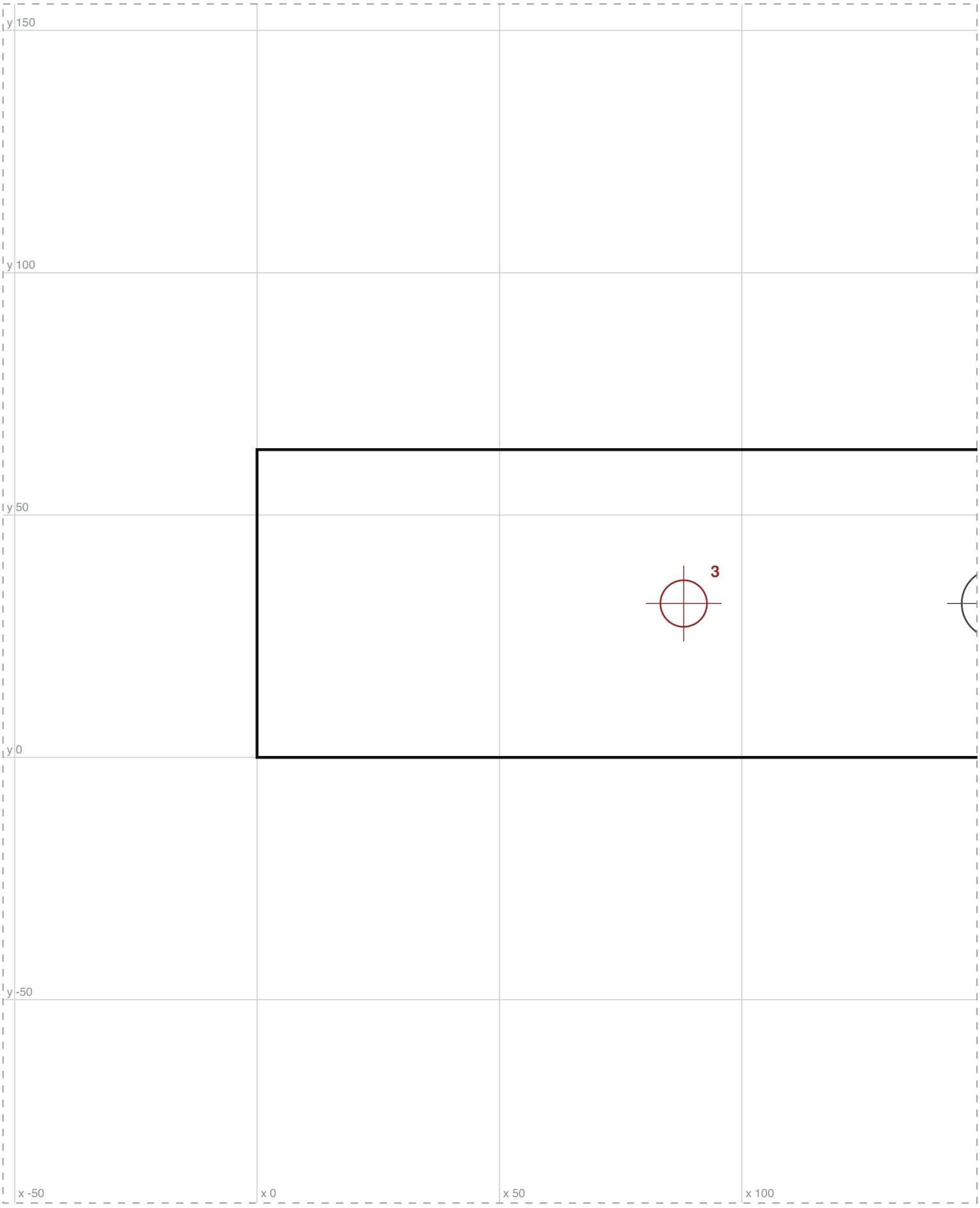


# Seat rear (x2: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 285 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 9.6 mm (3/8 in)

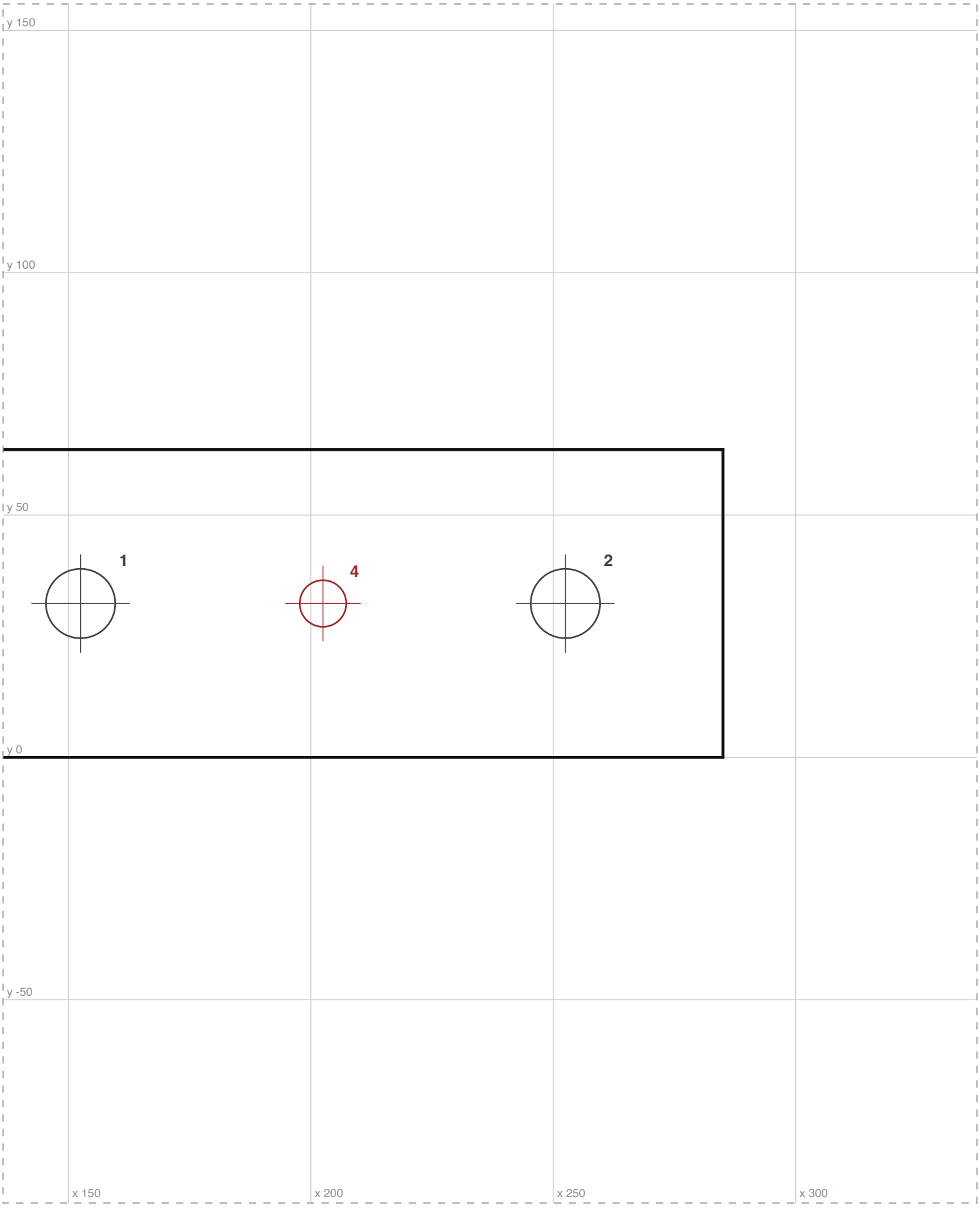


# Seat rear (x2: mark each)

page A2 · full size · Flat bar 3/8 x 2-1/2 in, 285 mm long · lay it on the part, x along its length from one end

50 mm: measure it

9.6 mm (3/8 in)  14.3 mm (9/16 in)

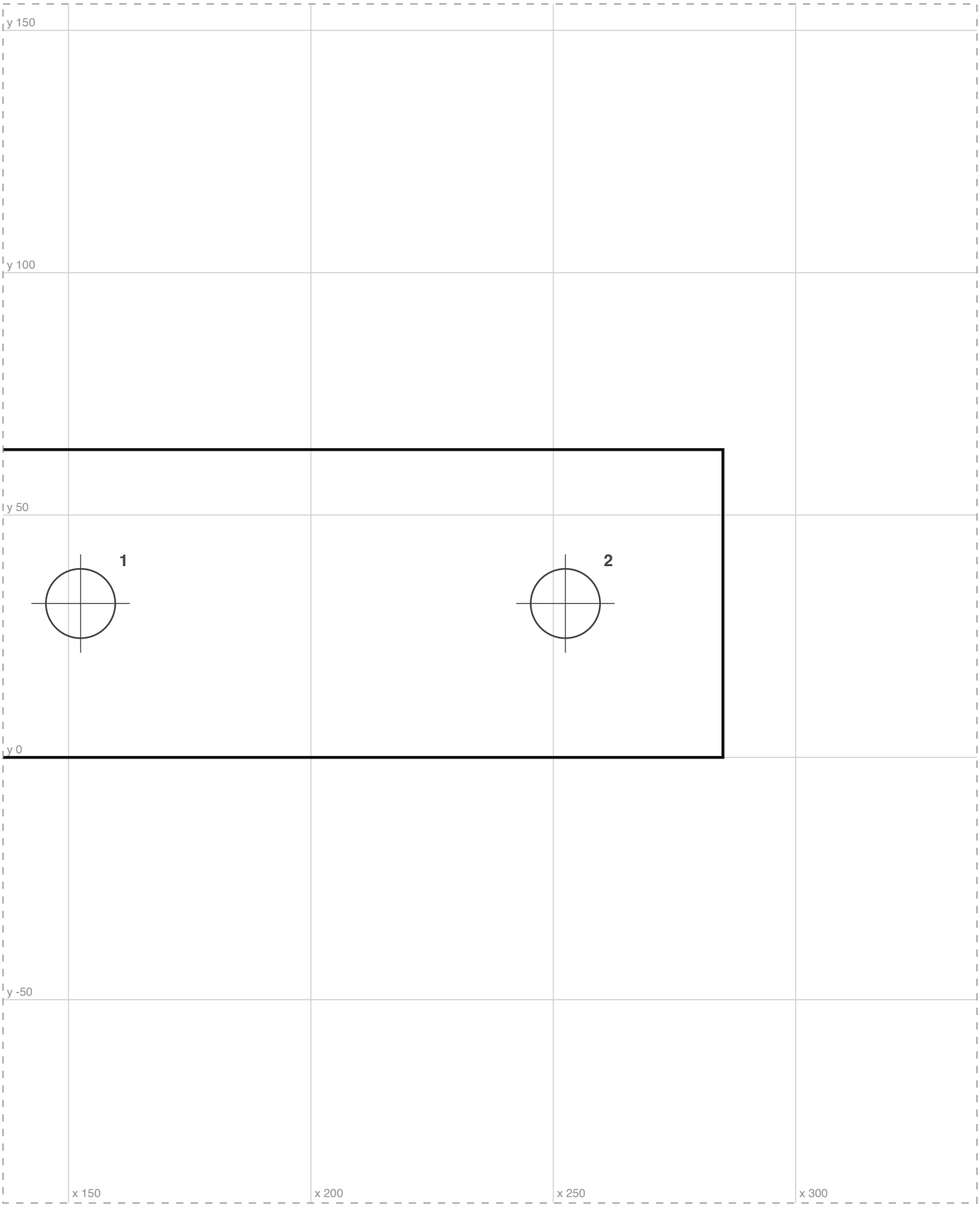


# Pedestal bar rear (x2: mark each)

page A2 · full size · Flat bar 3/8 x 2-1/2 in, 285 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 14.3 mm (9/16 in)

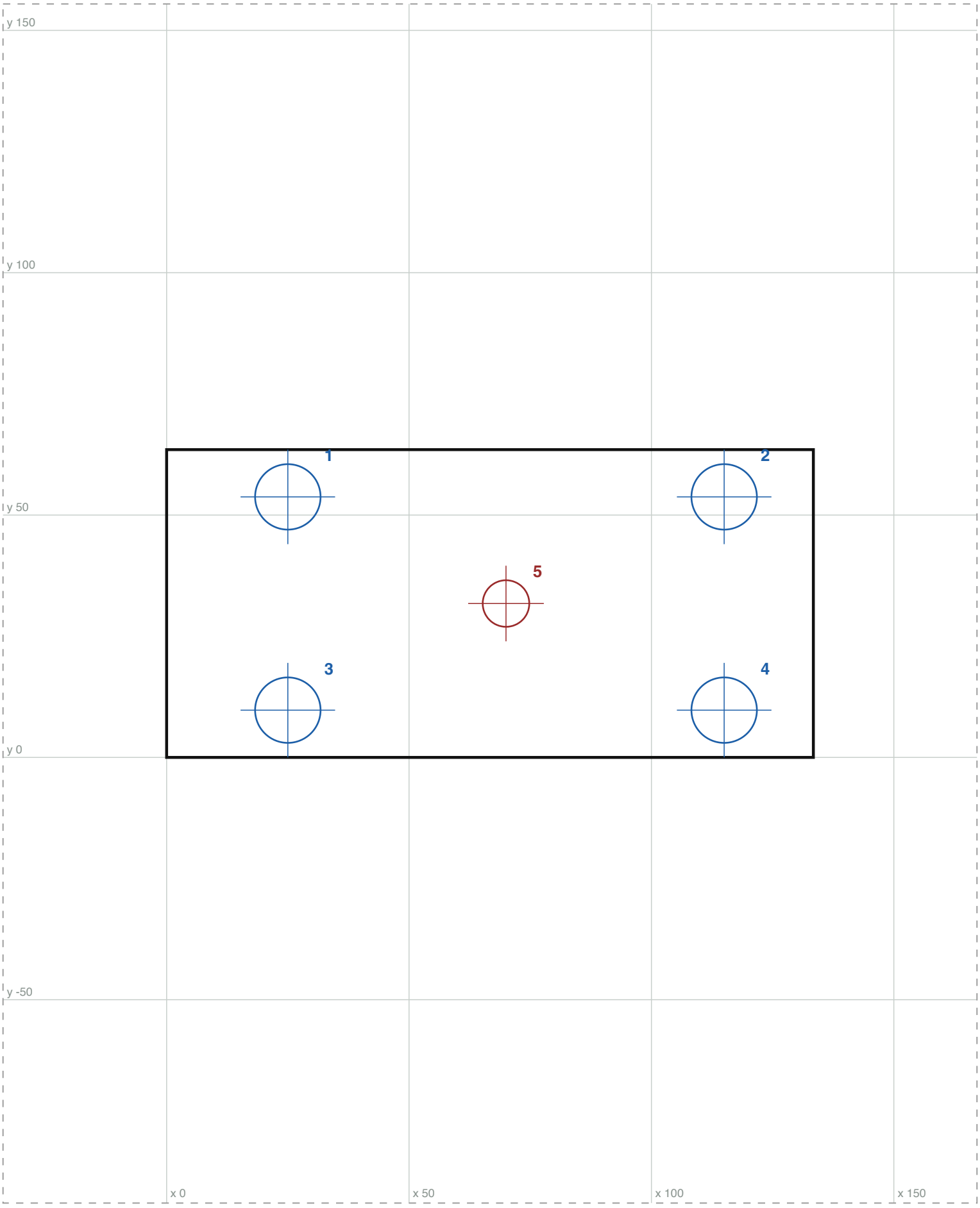


# Beam seat (x2: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 133 mm long · lay it on the part, x along its length from one end

50 mm: measure it

9.6 mm (3/8 in)  13.5 mm (17/32 in)

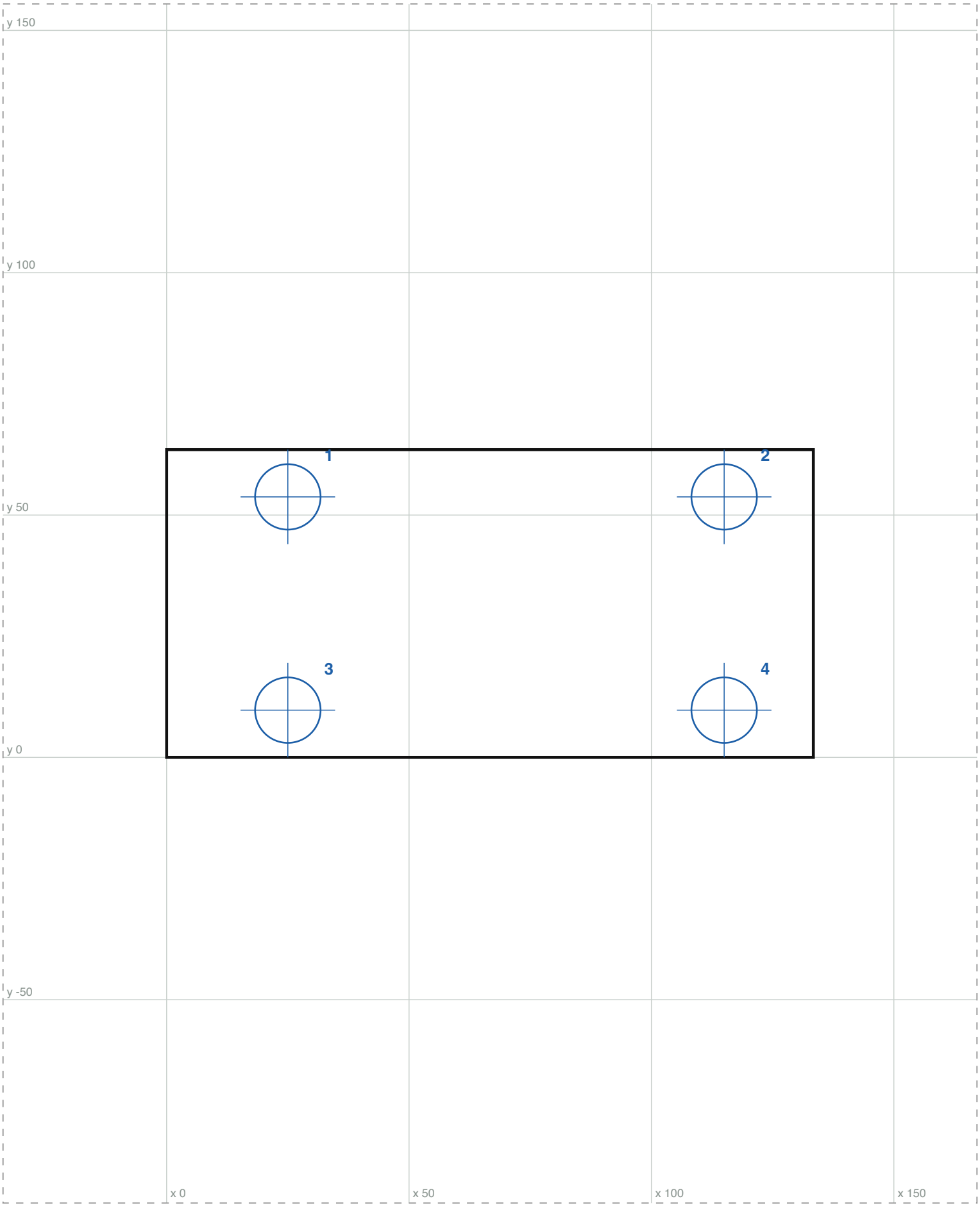


# Beam cap (x2: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 133 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 13.5 mm (17/32 in)

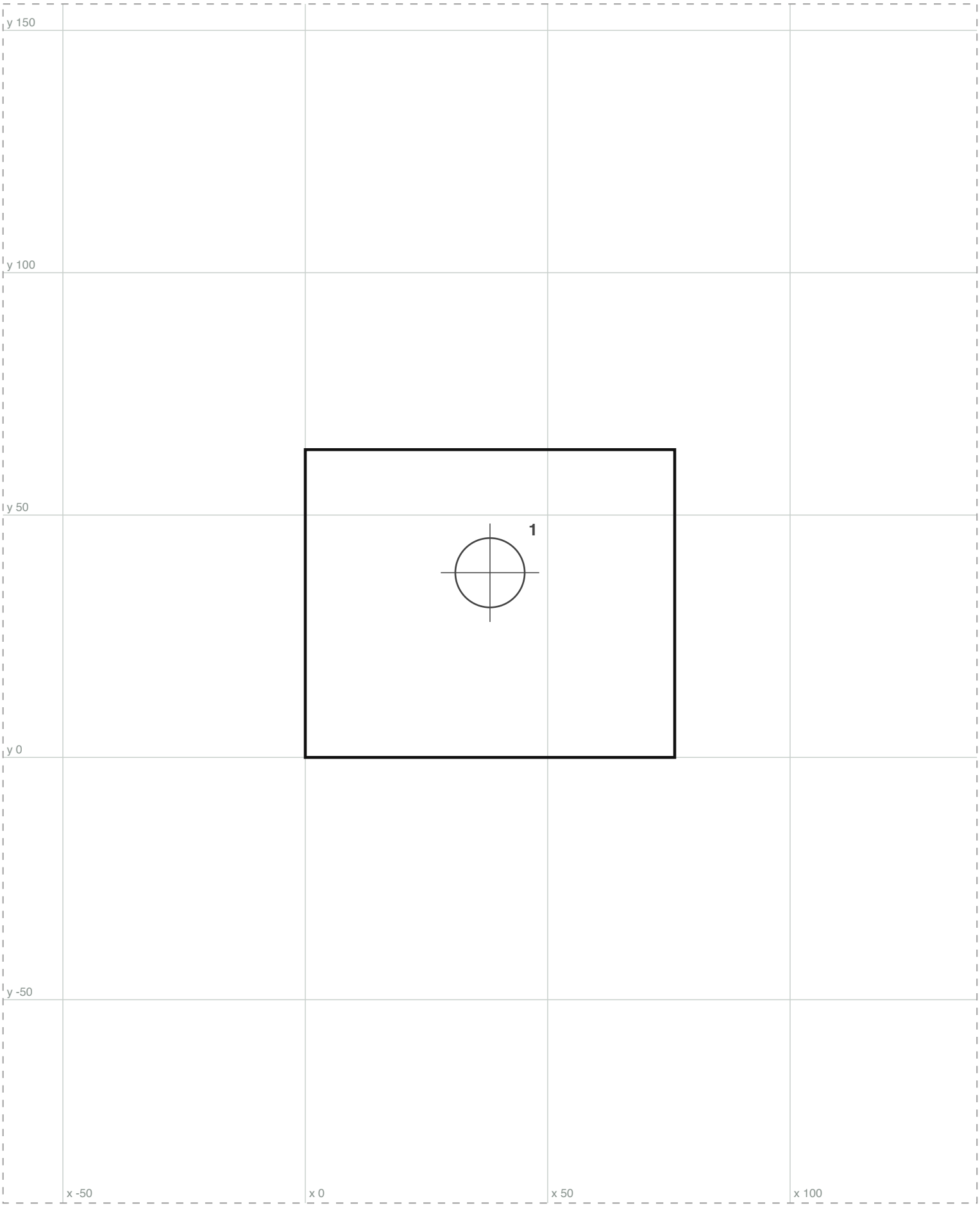


# Girder end (x4: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 76 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 14.3 mm (9/16 in)

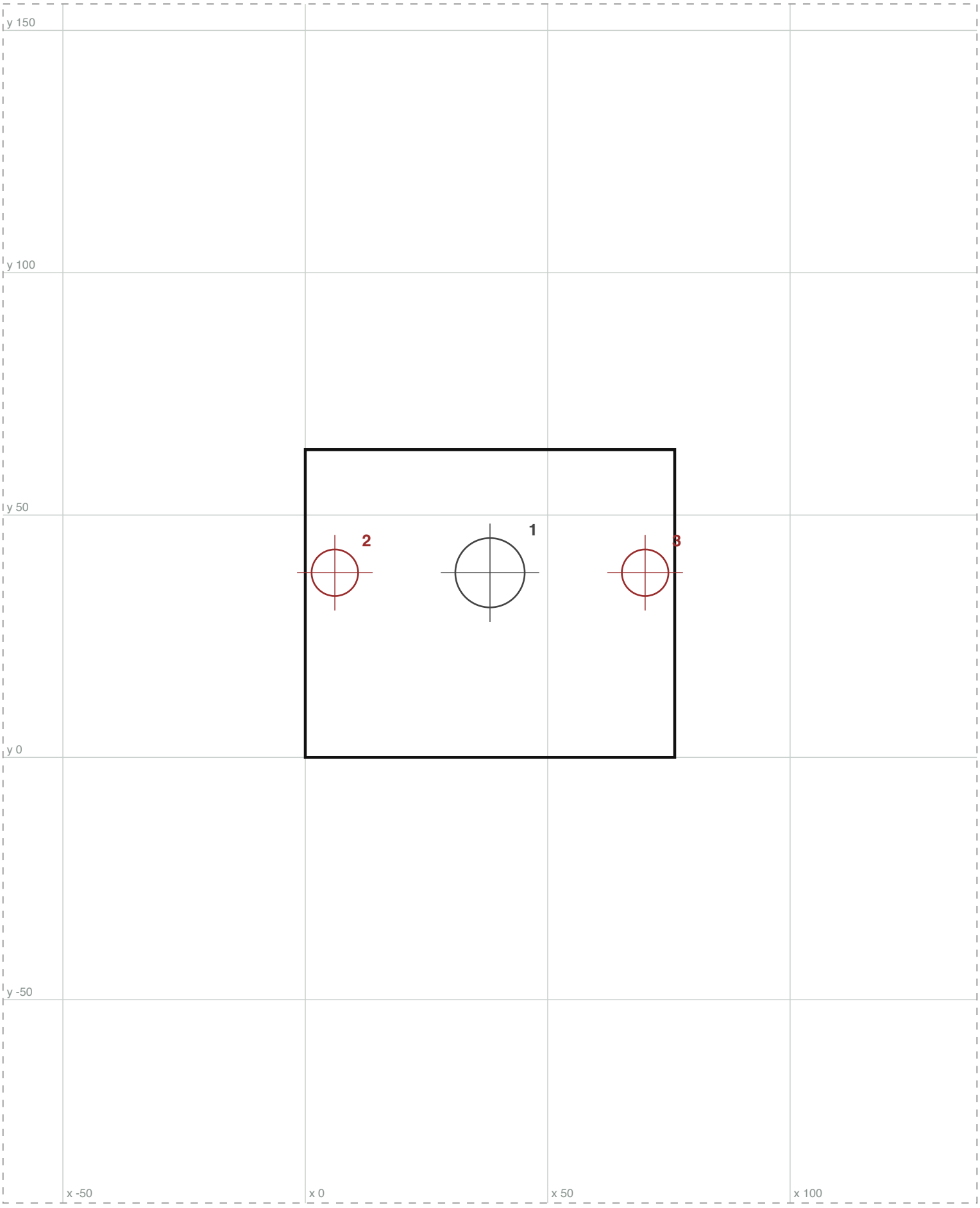


# Girder end split (x4: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 76 mm long · lay it on the part, x along its length from one end

50 mm: measure it

9.6 mm (3/8 in)  14.3 mm (9/16 in)

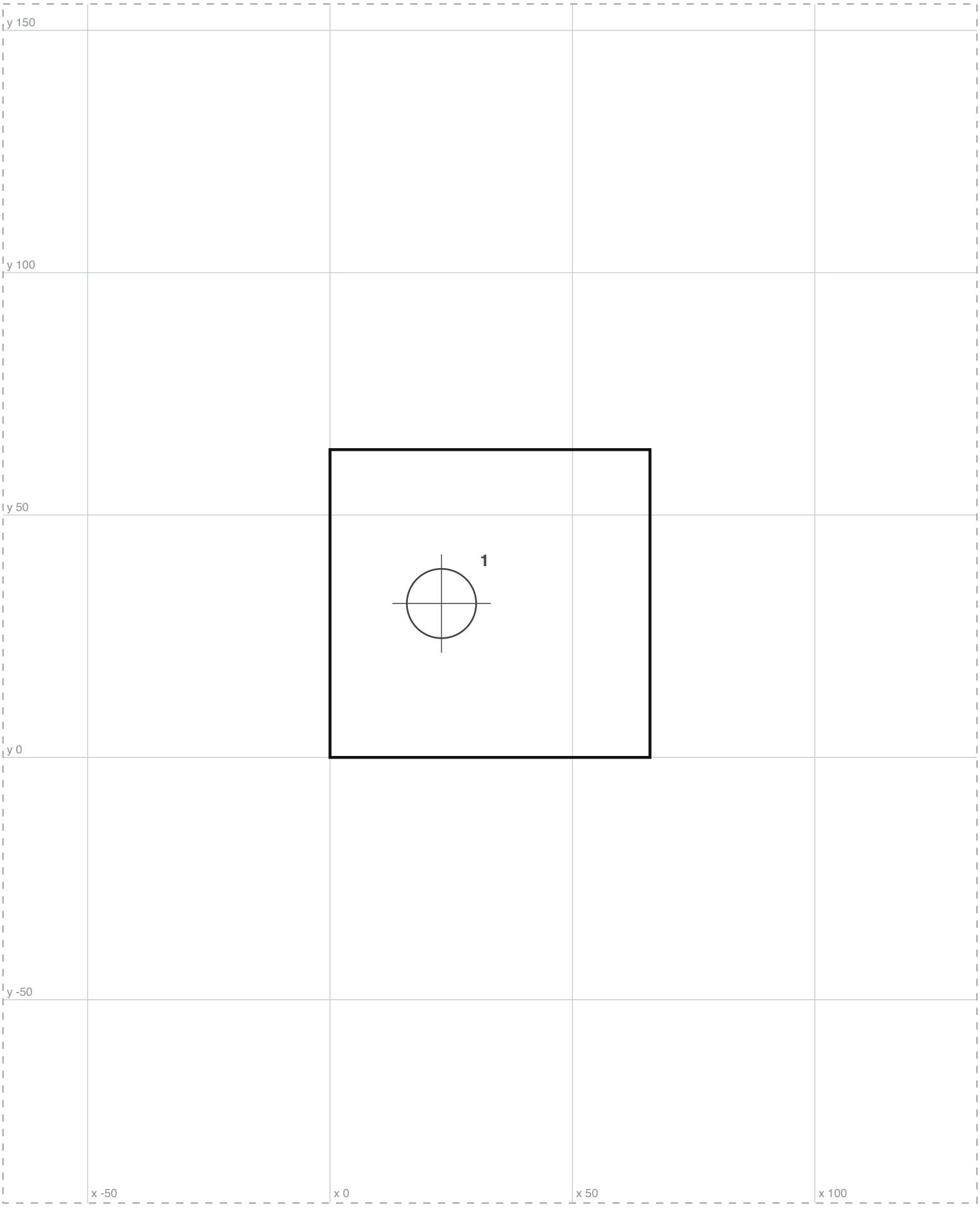


# Pedestal bar / seat front (x4: mark each)

page A1 · full size · Flat bar 3/8 x 2-1/2 in, 66 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 14.3 mm (9/16 in)

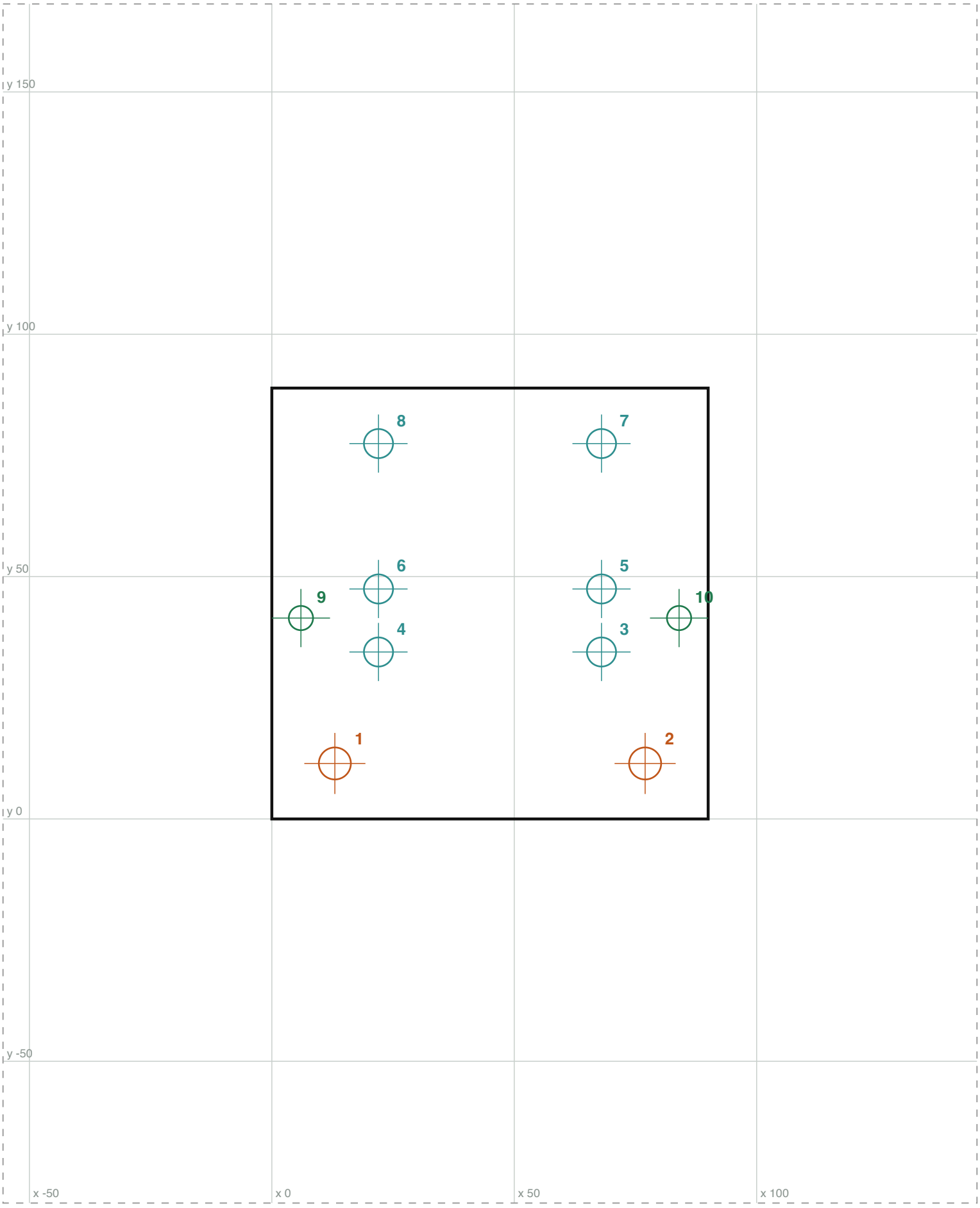


# Pad BK

page A1 · full size · Flat bar 3/8 x 3-1/2 in, 90 mm long · lay it on the part, x along its length from one end

50 mm: measure it

5 mm (M5 tap drill (4.2 mm), tap M5) 6 mm (M6 tap drill (5 mm), tap M6) 6.6 mm (17/64 in)

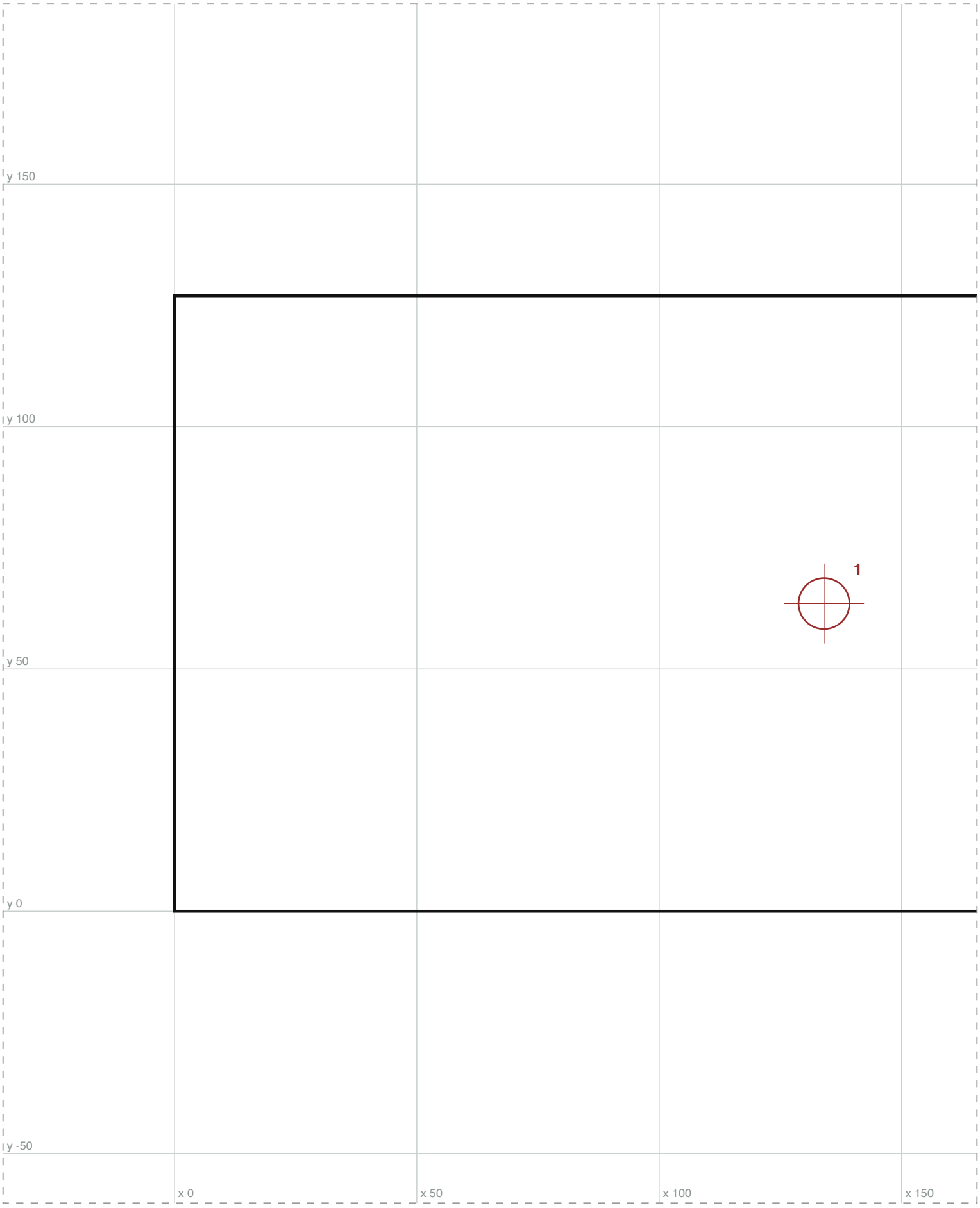


# Beam back plate

page A1 · full size · Flat bar 3/8 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 10.5 mm (13/32 in)

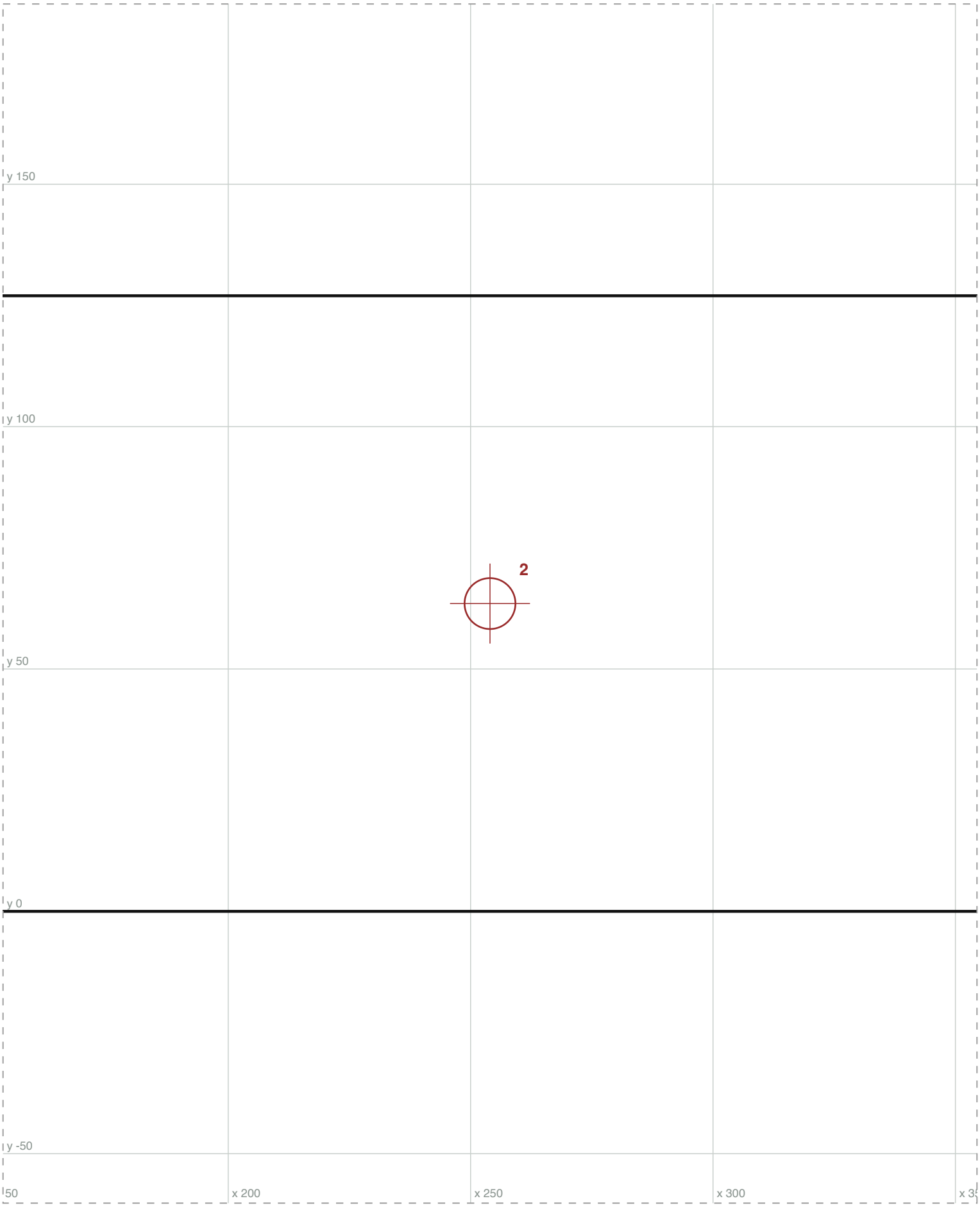


# Beam back plate

page A2 · full size · Flat bar 3/8 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 10.5 mm (13/32 in)

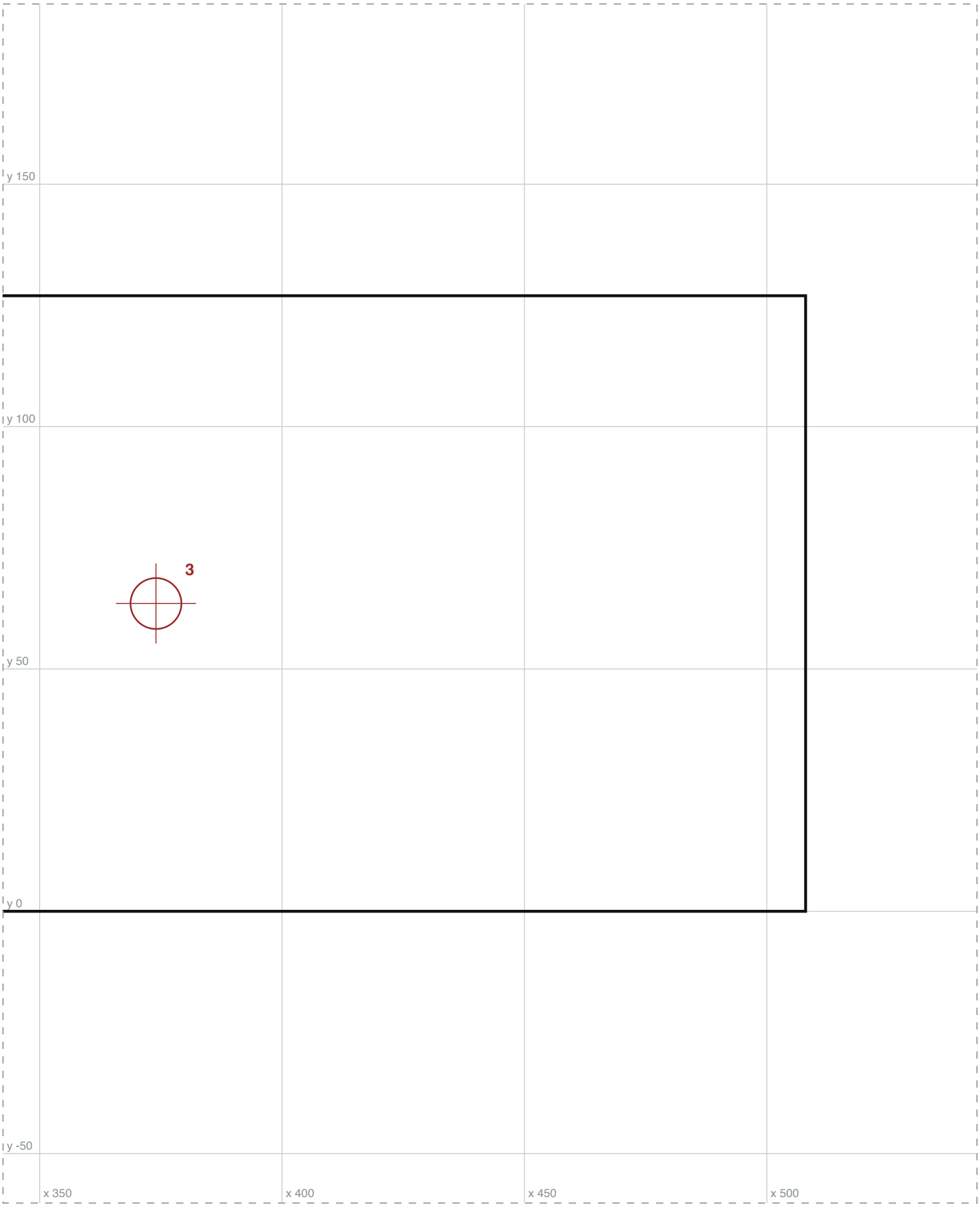


# Beam back plate

page A3 · full size · Flat bar 3/8 x 5 in, 508 mm long · lay it on the part, x along its length from one end

50 mm: measure it

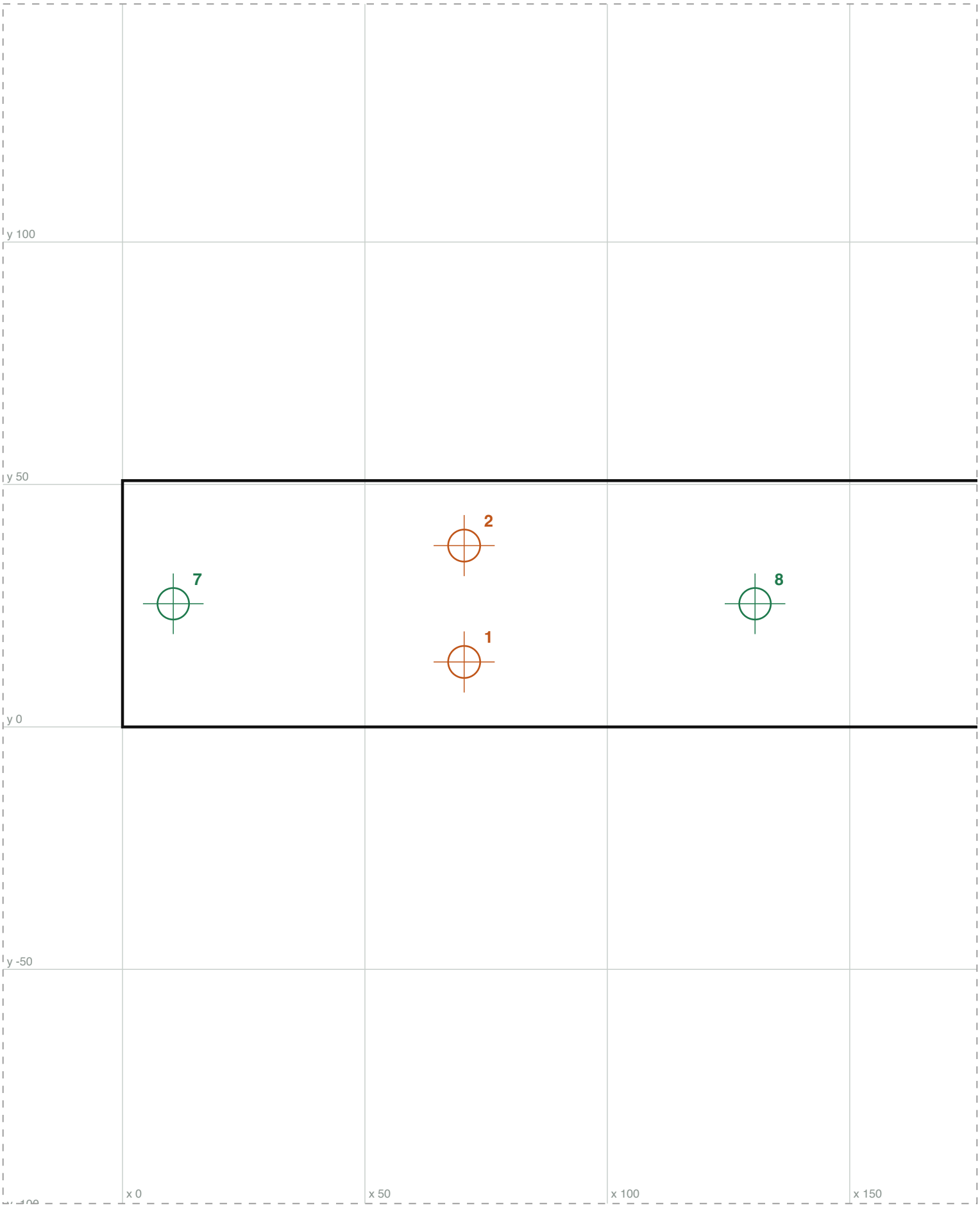
10.5 mm (13/32 in)



# Girder rear (z-face) (x2: mark each)

page A1 · full size · Rectangular tube 2 x 3 x 1/8 in, 340 mm long · lay it on the part, x along its length from one end50 mm: measure it

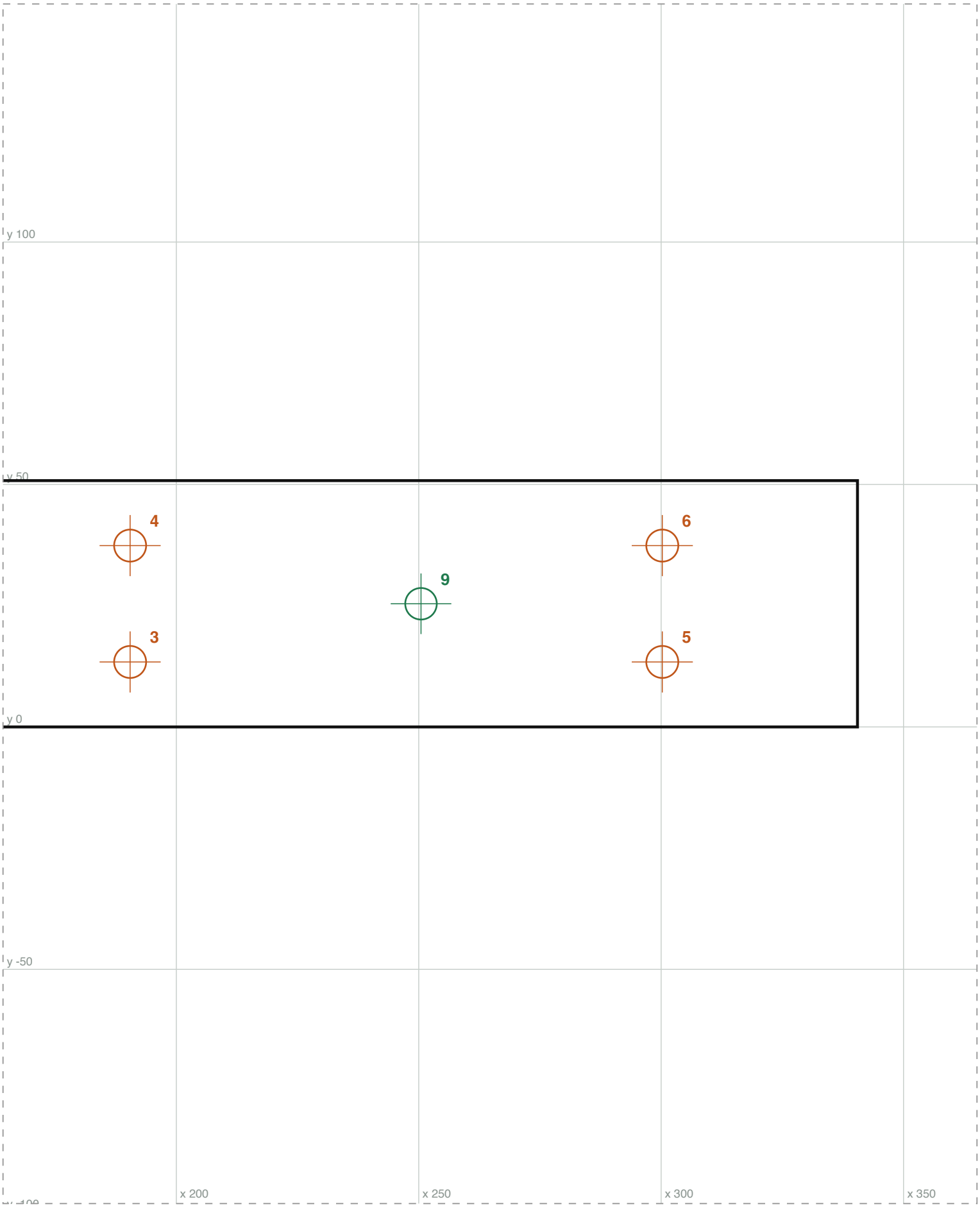
6.5 mm (1/4 in) 6.6 mm (17/64 in)



# Girder rear (z-face) (x2: mark each)

page A2 · full size · Rectangular tube 2 x 3 x 1/8 in, 340 mm long · lay it on the part, x along its length from one end50 mm: measure it

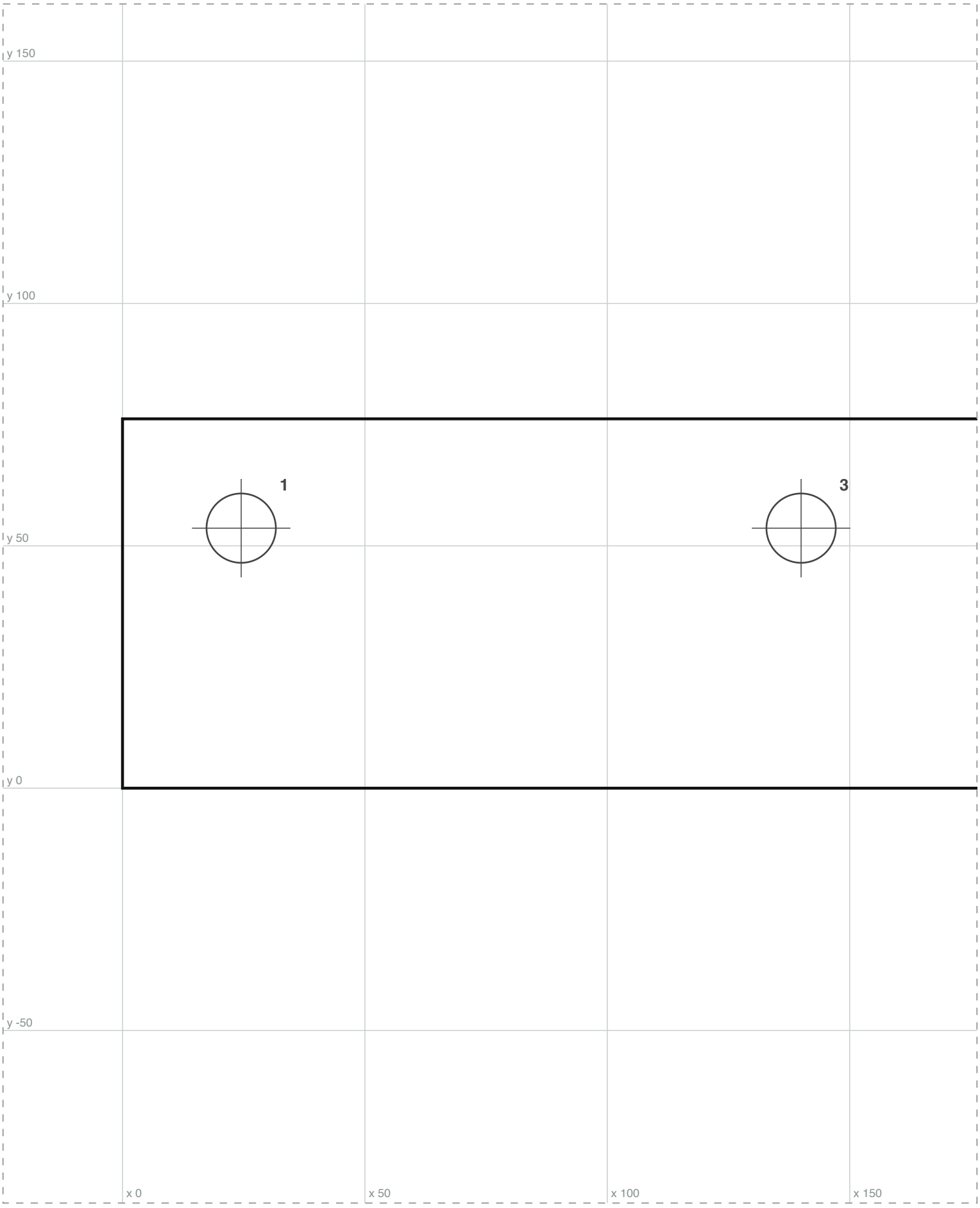
6.5 mm (1/4 in) 6.6 mm (17/64 in)



# Girder rear (x-face) (x2: mark each)

page A1 · full size · Rectangular tube 2 x 3 x 1/8 in, 340 mm long · lay it on the part, x along its length from one end50 mm: measure it

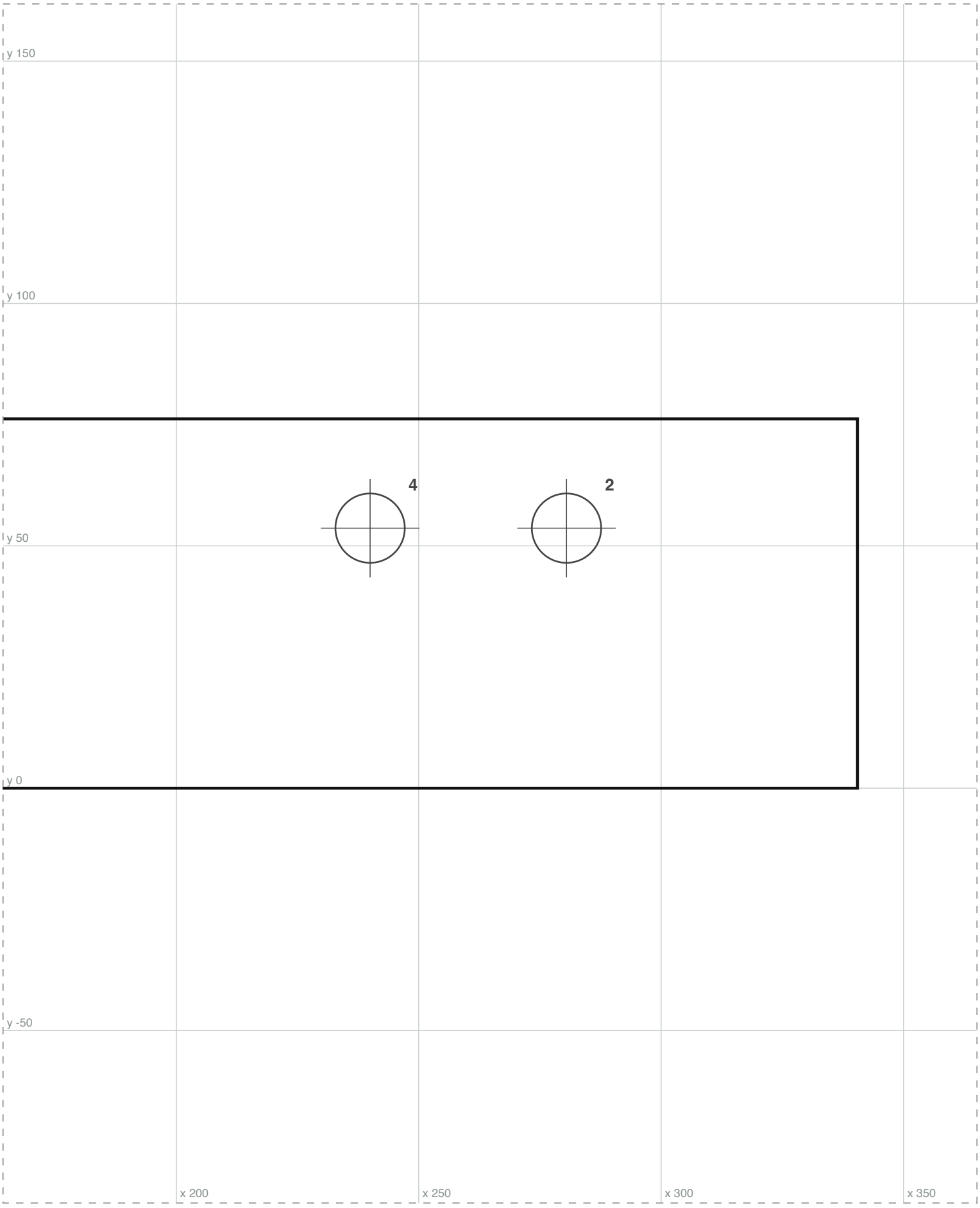
● 14.3 mm (9/16 in)



# Girder rear (x-face) (x2: mark each)

page A2 · full size · Rectangular tube 2 x 3 x 1/8 in, 340 mm long · lay it on the part, x along its length from one end50 mm: measure it

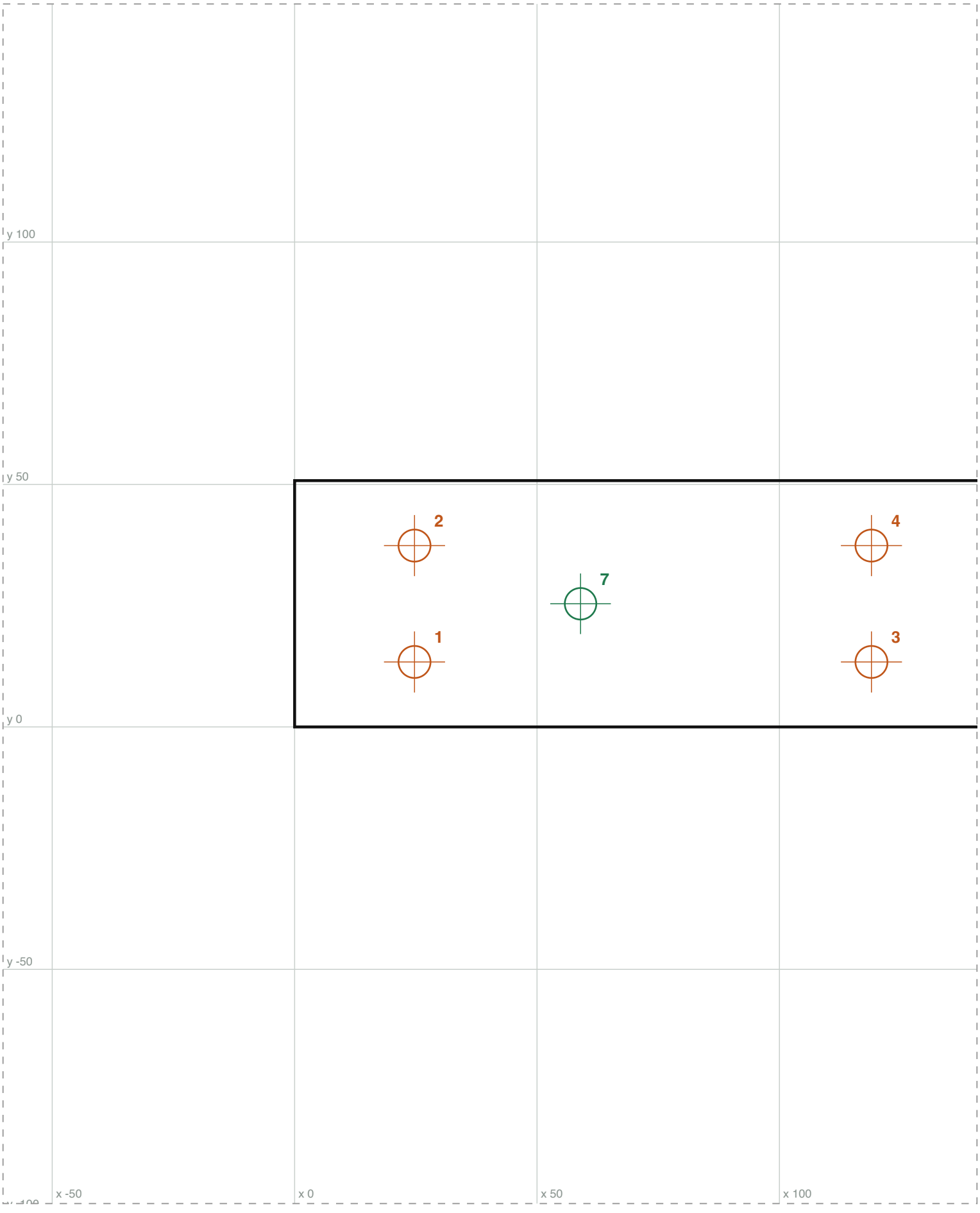
● 14.3 mm (9/16 in)



# Girder front (z-face) (x2: mark each)

page A1 · full size · Rectangular tube 2 x 3 x 1/8 in, 270 mm long · lay it on the part, x along its length from one end50 mm: measure it

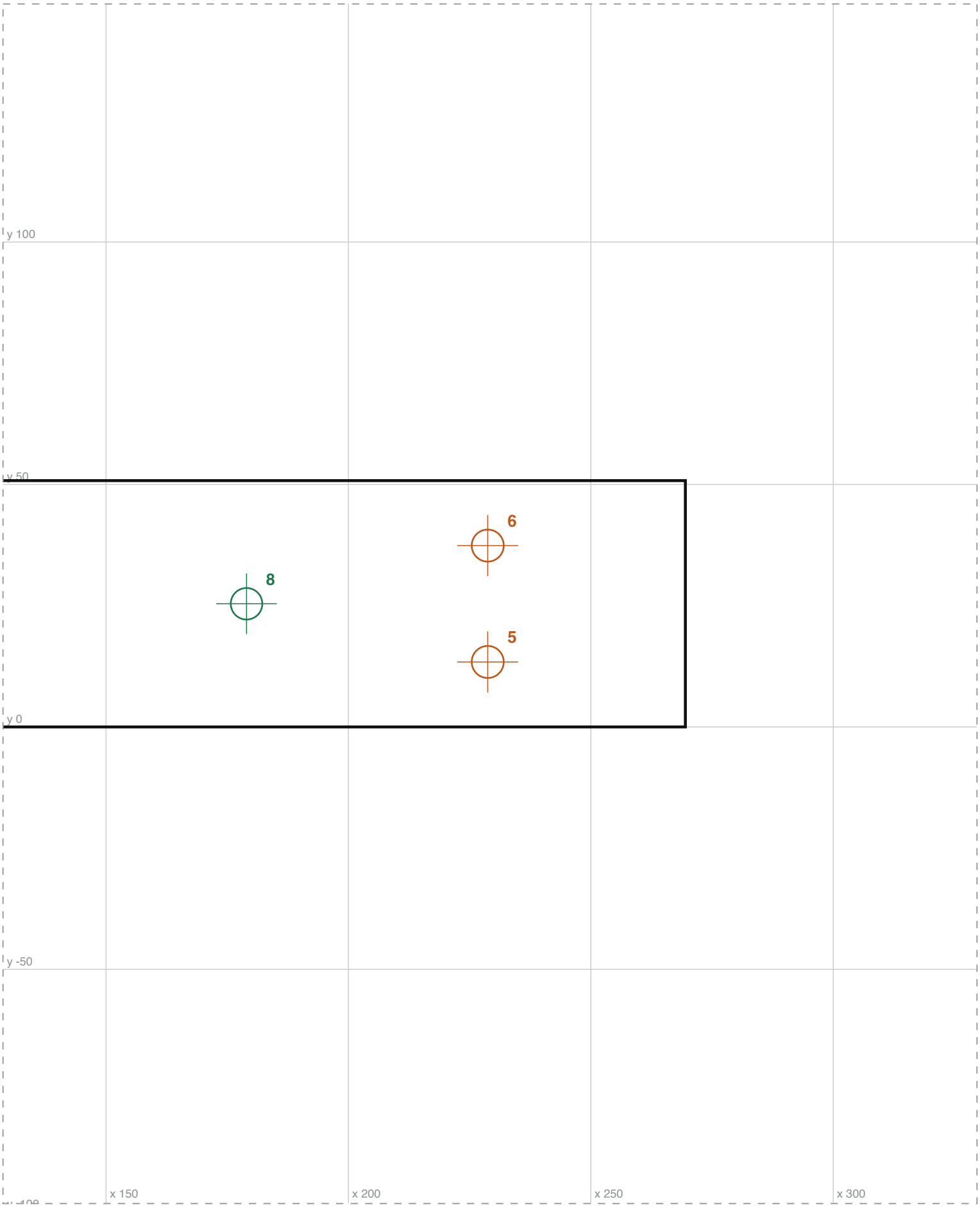
6.5 mm (1/4 in) 6.6 mm (17/64 in)



# Girder front (z-face) (x2: mark each)

page A2 · full size · Rectangular tube 2 x 3 x 1/8 in, 270 mm long · lay it on the part, x along its length from one end50 mm: measure it

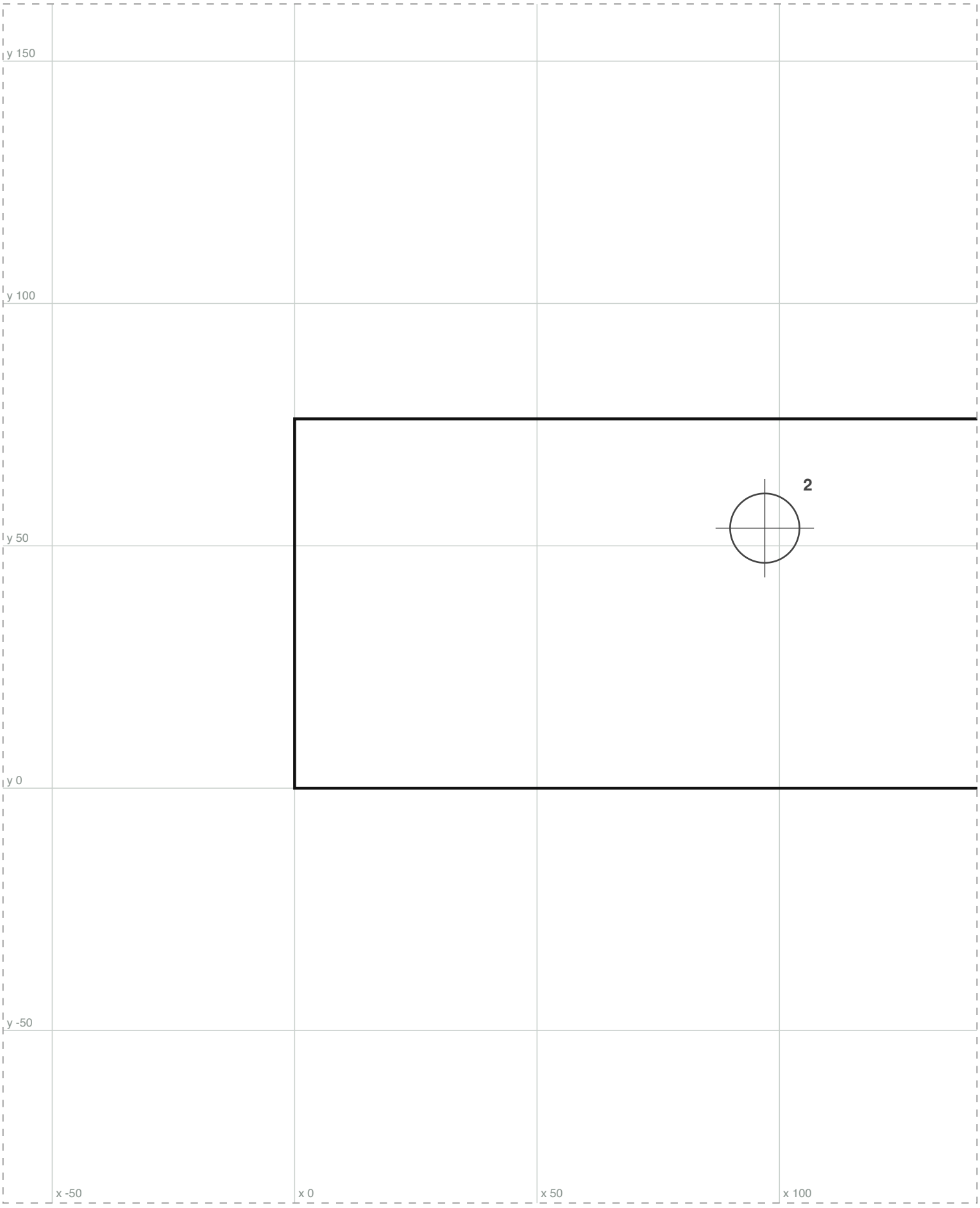
6.5 mm (1/4 in) 6.6 mm (17/64 in)



# Girder front (x-face) (x2: mark each)

page A1 · full size · Rectangular tube 2 x 3 x 1/8 in, 270 mm long · lay it on the part, x along its length from one end50 mm: measure it

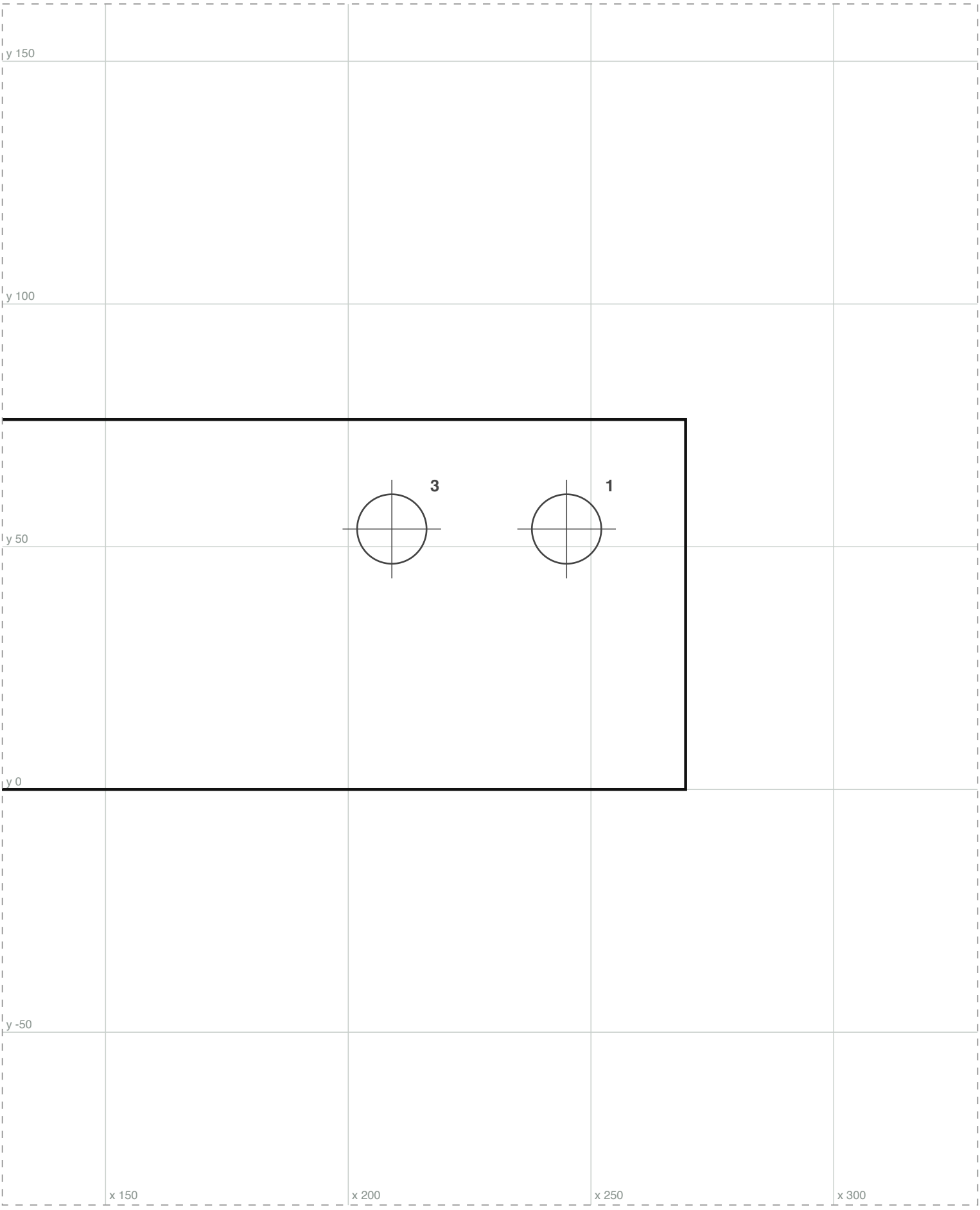
● 14.3 mm (9/16 in)



# Girder front (x-face) (x2: mark each)

page A2 · full size · Rectangular tube 2 x 3 x 1/8 in, 270 mm long · lay it on the part, x along its length from one end50 mm: measure it

● 14.3 mm (9/16 in)

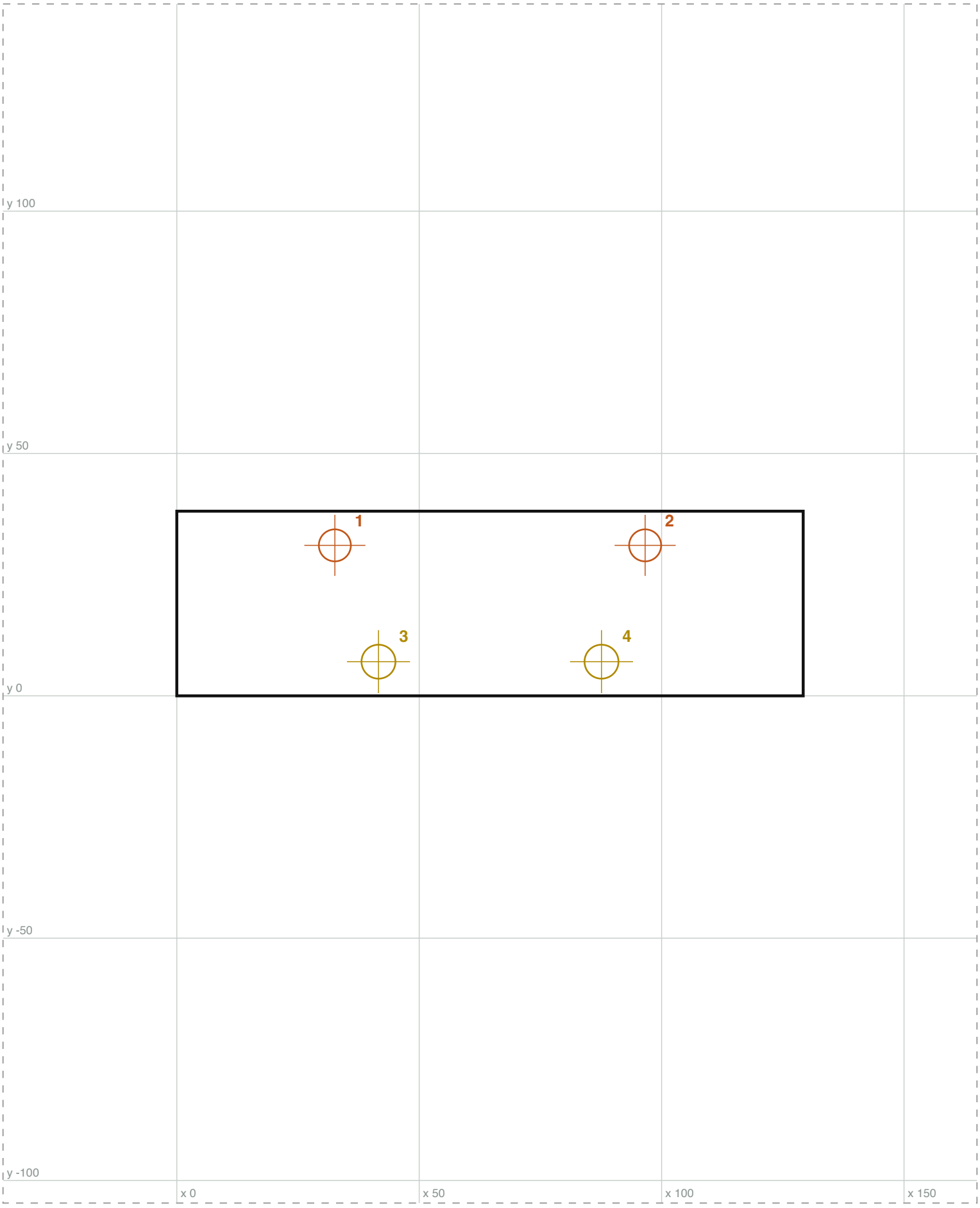


# Cross BF front (z-face)

page A1 · full size · Square tube 1-1/2 x 1-1/2 x 1/8 in, 129 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 6.6 mm (17/64 in) ● 7 mm (9/32 in)

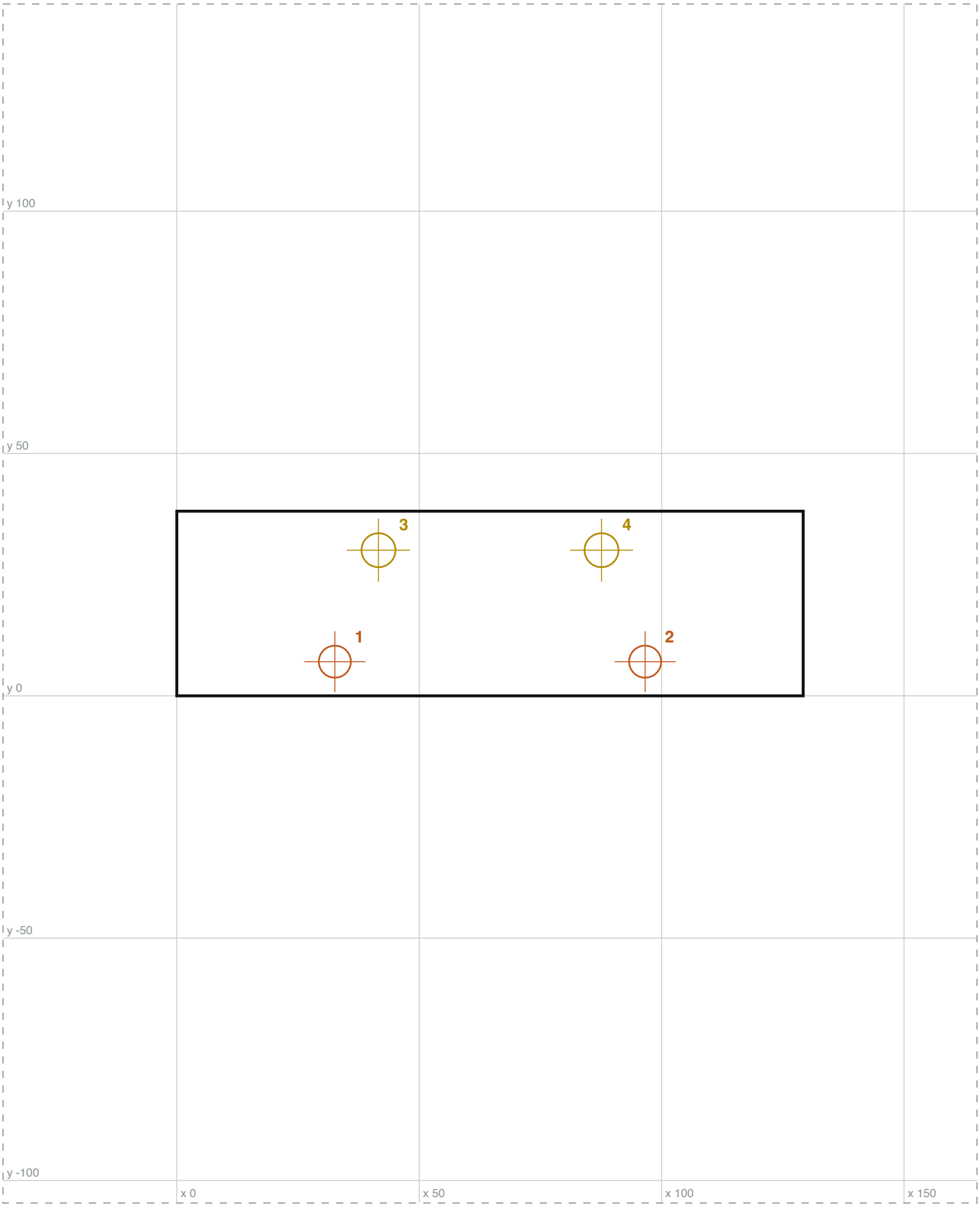


# Cross BK rear (z-face)

page A1 · full size · Square tube 1-1/2 x 1-1/2 x 1/8 in, 129 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 6.6 mm (17/64 in) ● 7 mm (9/32 in)



# Pedestal column (x-face) (x6: mark each)

page A1 · full size · Square tube 2 x 2 x 1/8 in, 91 mm long · lay it on the part, x along its length from one end

50 mm: measure it

● 14.3 mm (9/16 in)

